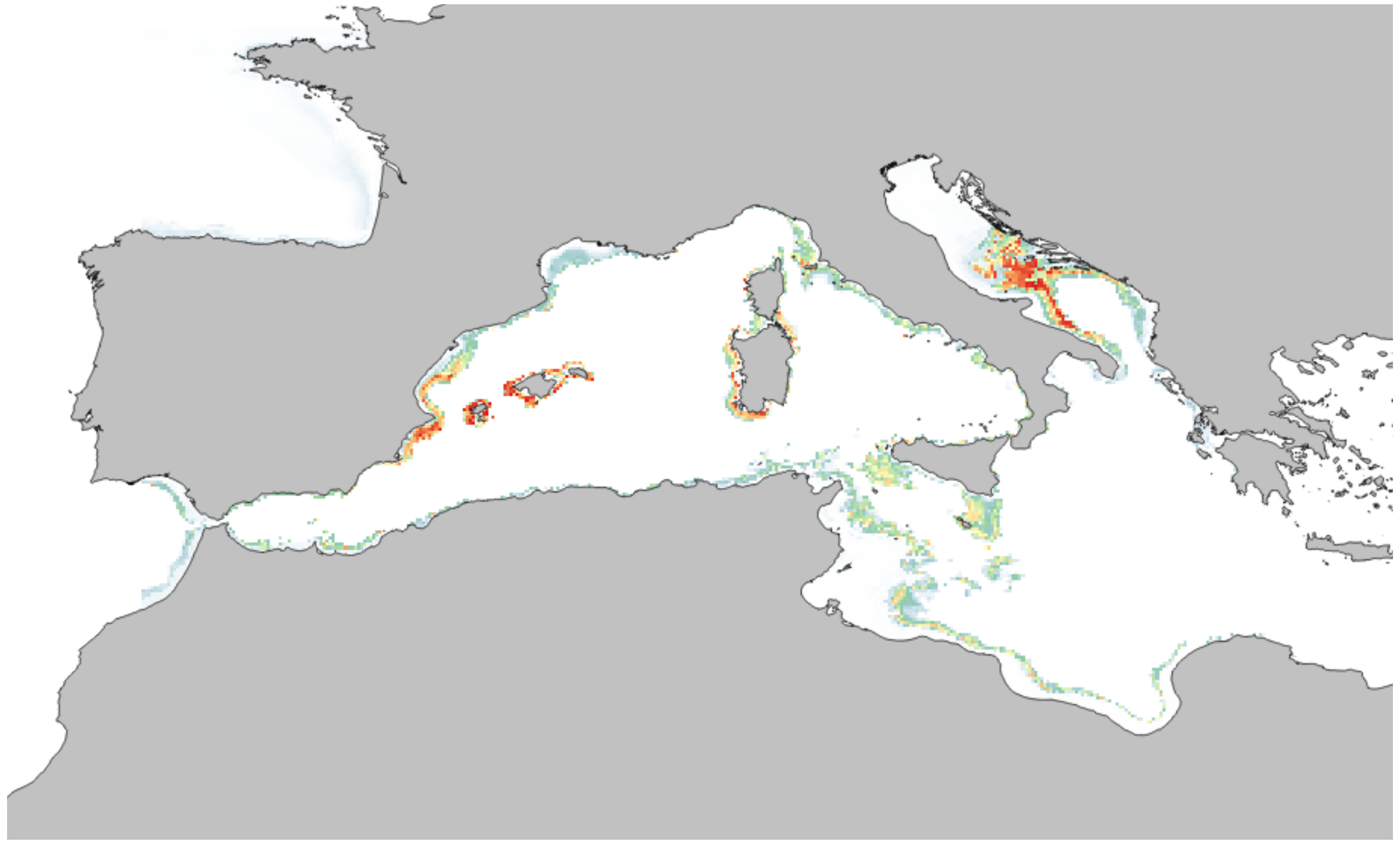
Evaluating, Transferring, and Ensembling

Modeling the distribution of marine biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Measuring Predictive Performance



Model evaluation / performance

Key goals:

- **Verify consistency:** do predictions match observations (presences and absences)?
- **Assess transferability:** how well does the model generalize to new data?
- **Evaluate ecological realism:** does the model make biological sense?

Fundamental question: is the model suitable for the purpose (e.g., impact assessment)?

Model evaluation

By comparing predictions with observations we assess for **‘false positives’** - model predicts occurrence where the species is absent - and **‘false negatives’** - model predicts absent where the species is present.This information can be summarized in confusion matrices.

Confusion matrix

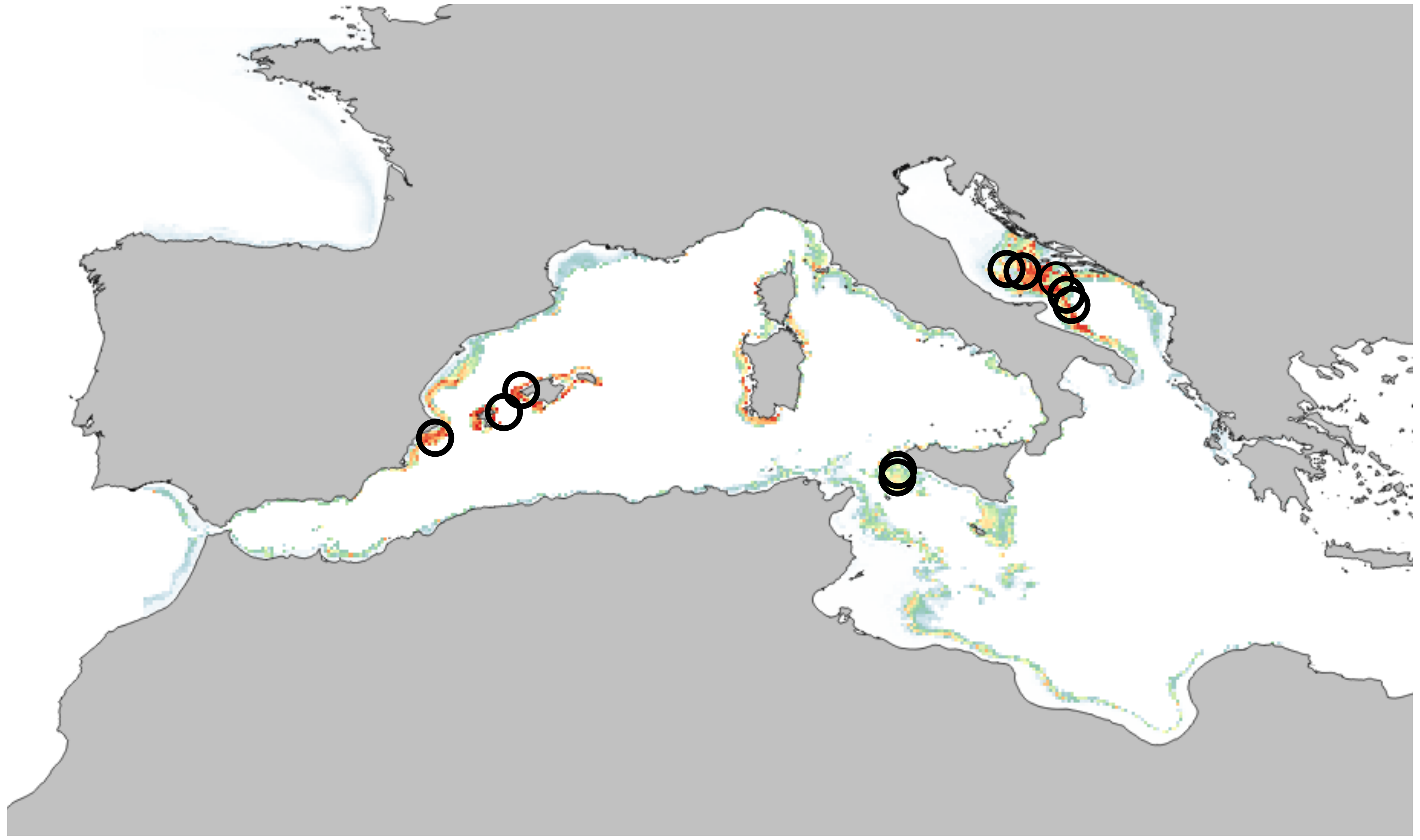
Types of prediction errors

		Observation	
		Presence	Absence
Prediction	Presence	True Positive	False Positive
	Absence	False Negative	True Negative

Perfect models only retrieve true positives and true negatives.

Predictions are continuous surfaces (e.g., probability, from 0 to 1).

One cannot compare observations (e.g., 1s) with the corresponding model output (e.g., P: 0.7).



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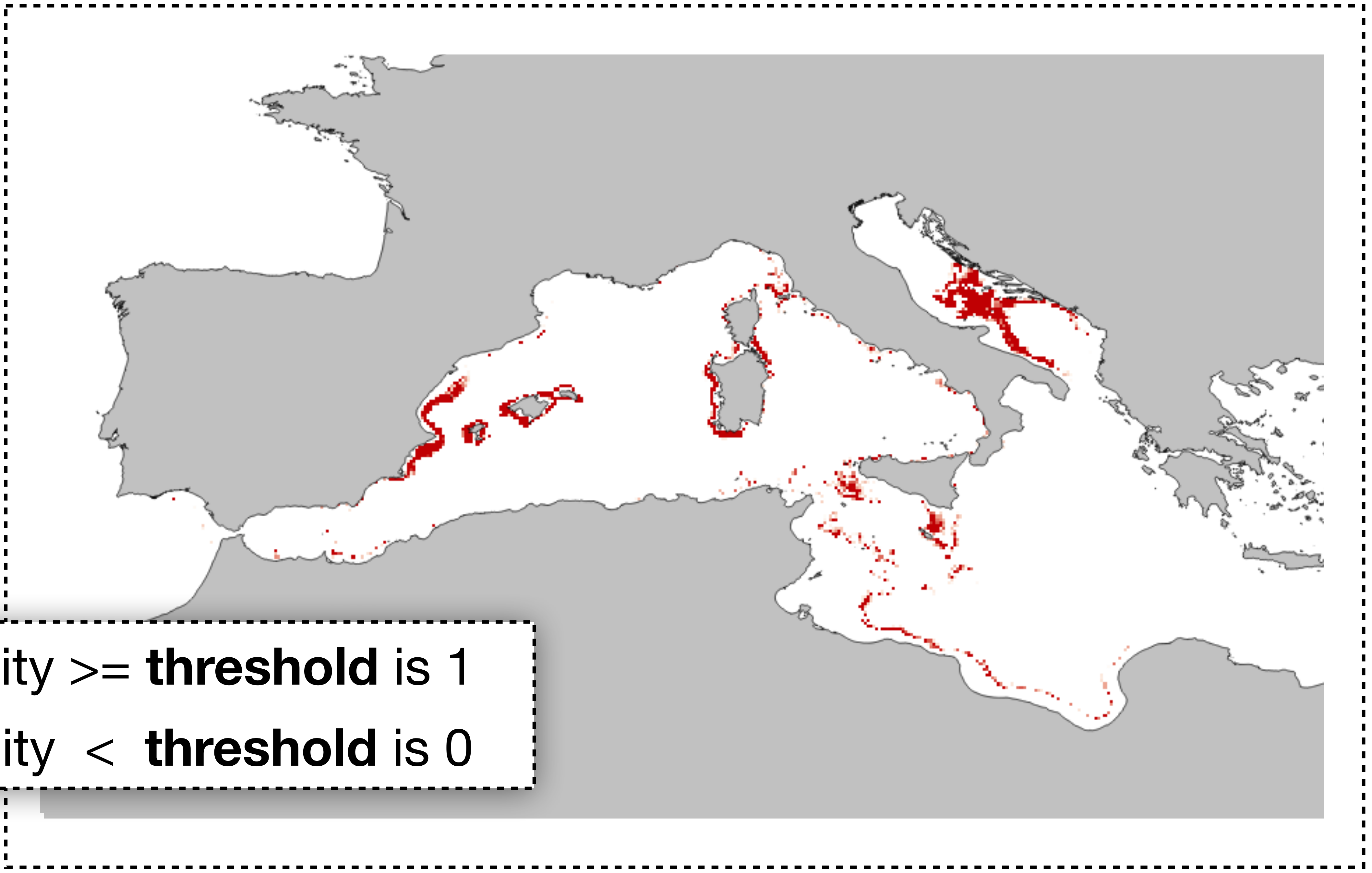


Probability >= **threshold** is 1

Probability < **threshold** is 0

Predictions need to be reclassified as binary responses (presence-absence) for comparison with the observed data.

Evaluation metrics: threshold-dependent

The elements of the confusion matrix are used to compute various evaluation criteria that measure the performance of the model, **after applying a threshold to predictions:**

Sensitivity: proportion of presences correctly predicted;

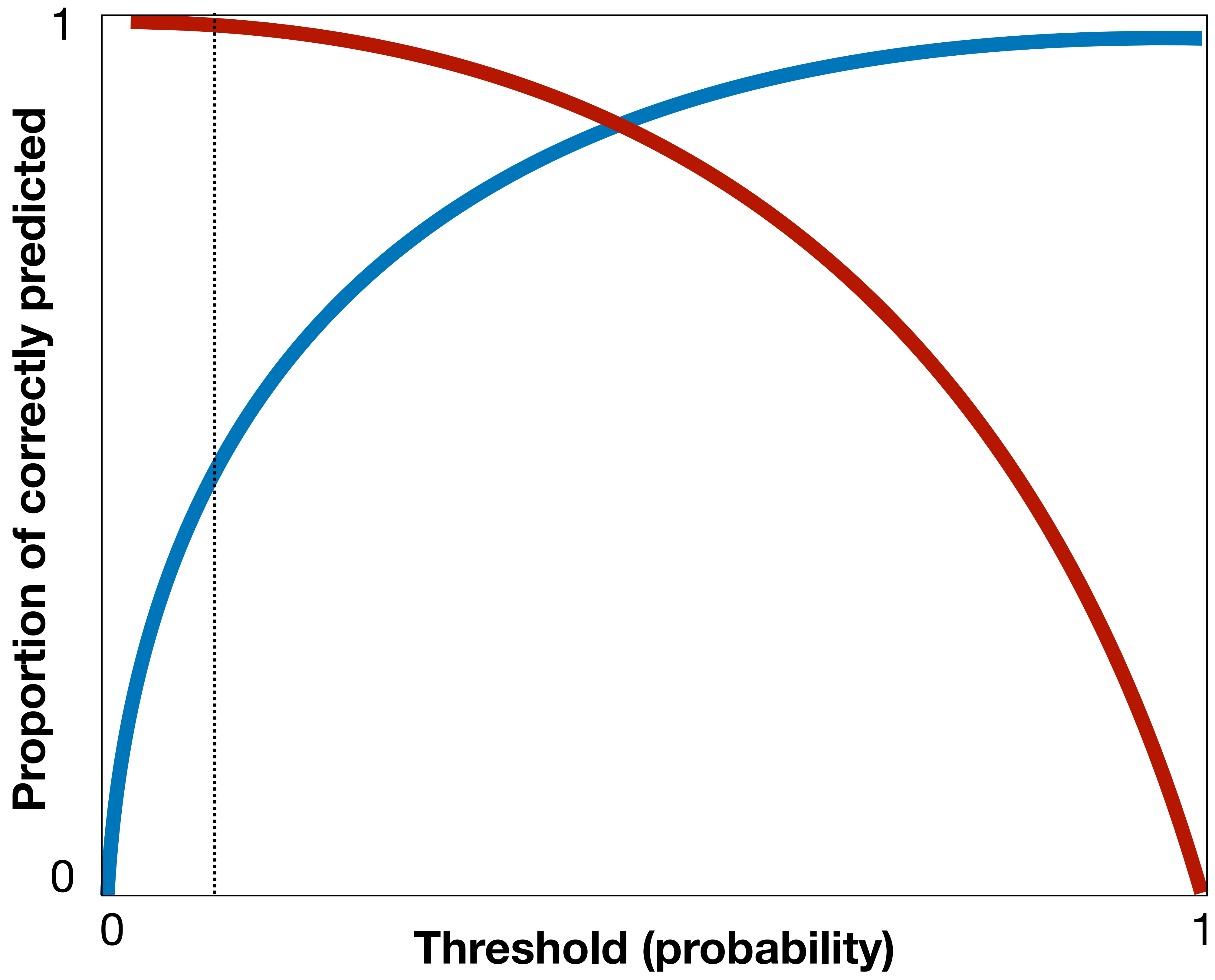
Specificity: proportion of absences correctly predicted;

True Skill Statistics: Sensitivity + Specificity - 1

TSS describes how well the model predicts presences and absences;

TSS ranges from -1 to +1 (0 <= no better than a random model).

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.



Probability >= **threshold** is 1

Probability < **threshold** is 0

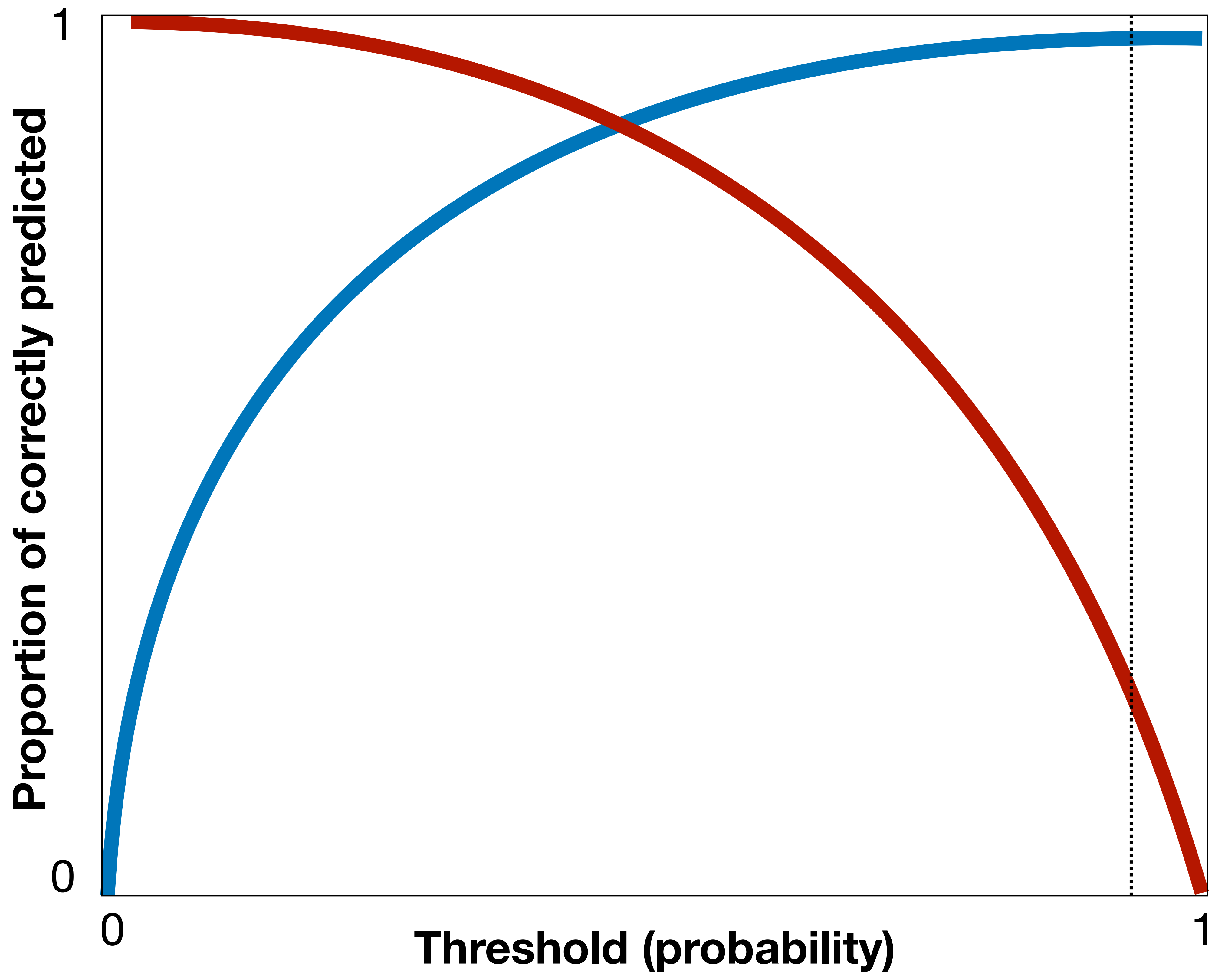
Effect on the indices

Sensitivity (presences)

Specificity (absences)

Using a low threshold most cells (in the map) will return 1.

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.



Probability >= **threshold** is 1

Probability < **threshold** is 0

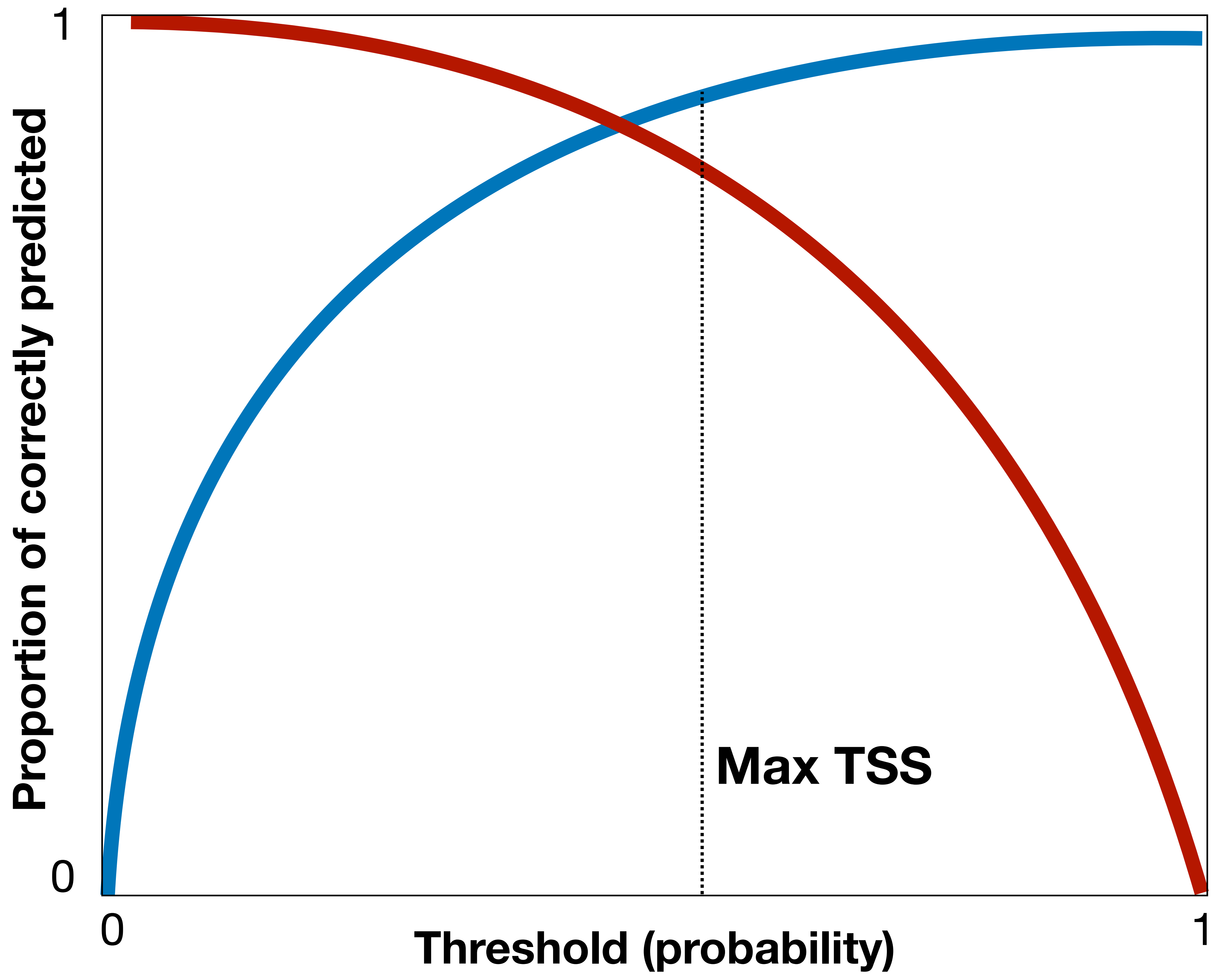
Effect on the indices

Sensitivity (presences)

Specificity (absences)

Using a high threshold most cells (in the map) will return 0.

There are threshold rules to maximize the agreement between observed data and the predicted binary surface.



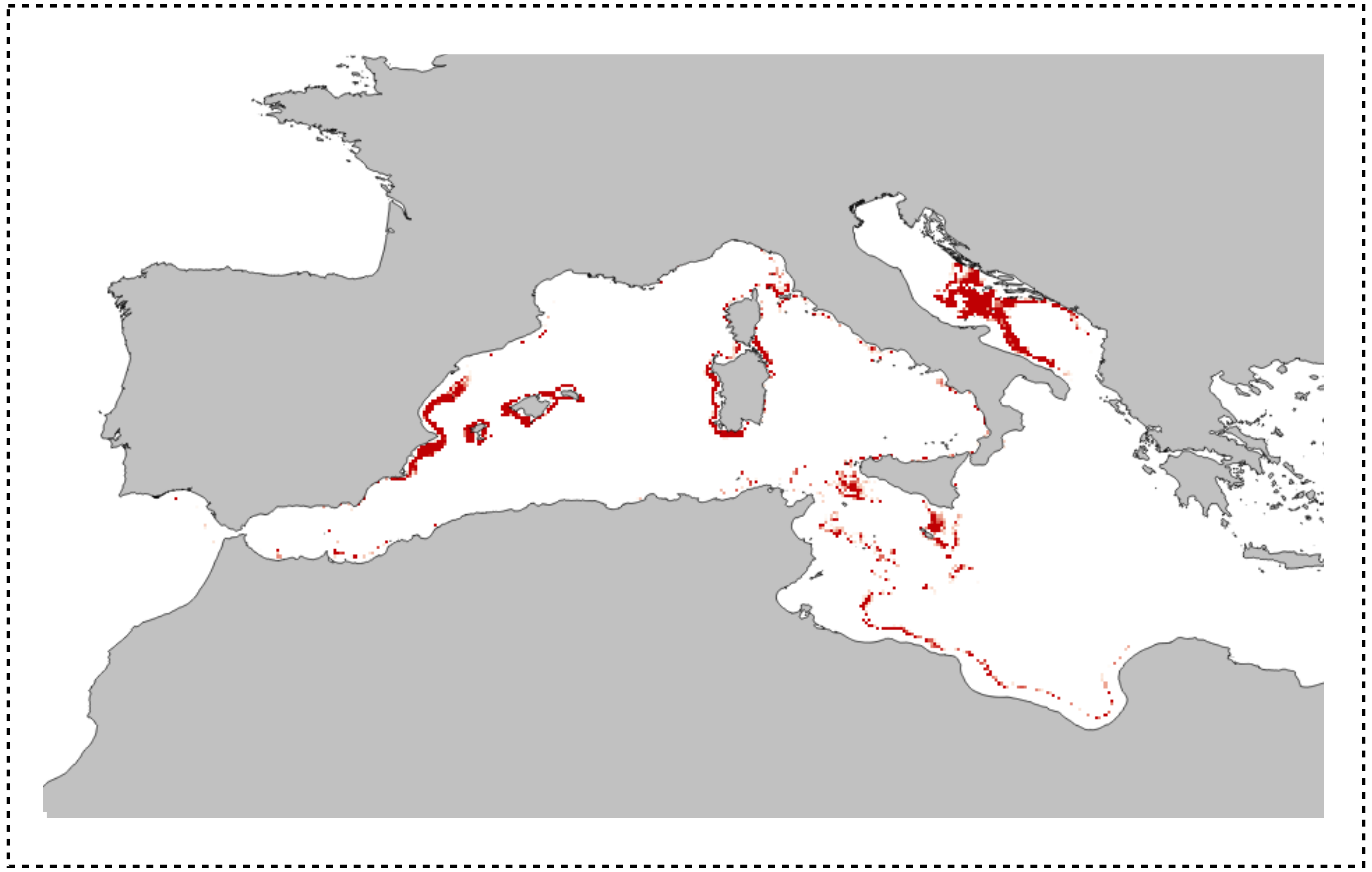
Probability >= **threshold** is 1

Probability < **threshold** is 0

Effect on the indices

- **Sensitivity (presences)**
- **Specificity (absences)**

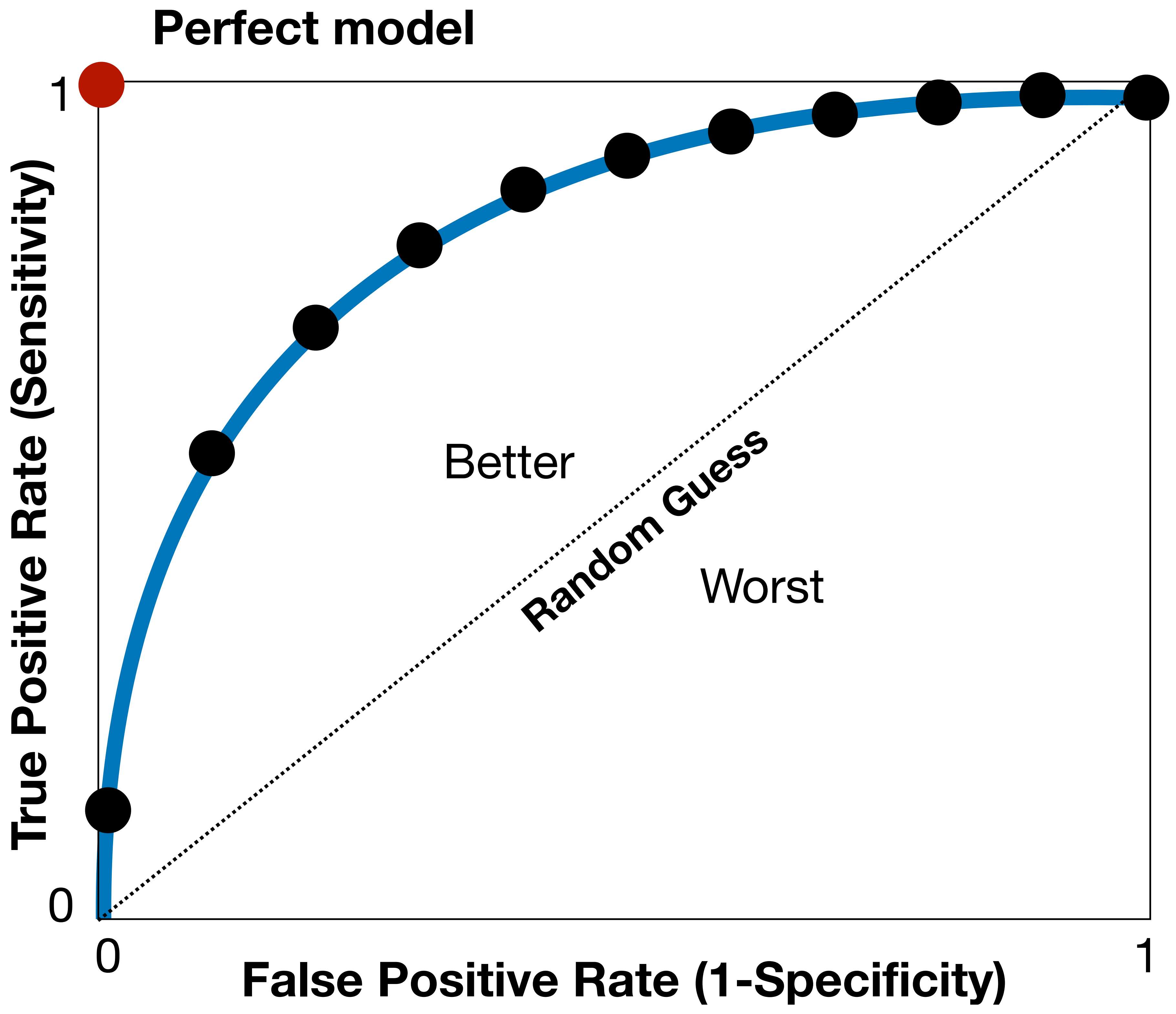
(2) **Maximization of sensitivity + specificity**



TSS = 0.93

Threshold-dependent measures of accuracy

Thresholds like “Max. TSS” allow **assessing the agreement between observed data and the model output** and **converting probability maps into binary maps.**



Threshold-independent evaluation criteria

Area **U**nder the **C**urve of the Receiver Operating Characteristic Plot.

Compared **true positive rate (sensitivity)** against **false positive rate (1-specificity)** across the range of all possible thresholds. The closer the curve from y-axis, the larger the **AUC**, and thus the more accurate the model.

AUC performance

Accuracy indices like AUC can be interpreted as follow:

1 - 0.9 : excellent model

0.9 - 0.8 : good model

0.8 - 0.7 : fair model

0.5 - 0.7 : poor model

0 - 0.5 : random model

The Flaw of Standard ROC-AUC: While the ROC-AUC is the most popular metric, it can be highly misleading when dealing with imbalanced datasets - a very common scenario in SDMs where you might have 100 presence records but 10,000 background points.

Why it Fails: ROC-AUC relies on Specificity. With a massive amount of background points, the impact of few occurrence records (presences) wrongly classified won't significantly impact the score, artificially inflating the model's perceived performance.

To solve this, we can use the Precision-Recall Curve and the Boyce index, which completely ignore Specificity and focuses on the minority class (Sensitivity).

The Precision-Recall Alternative: When the model predicts a presence, how often is it actually correct? AND Out of all actual presences, how many did the model successfully find? The Area Under the Precision-Recall Gain Curve (AUPRG) ranges from 0 to 1 (0 = no better than a random model).

The Boyce index: Tests (by correlation) if higher suitability predictions correspond to higher densities of presence points, which is the desired model behavior. Ranges from -1 to +1 (0 <= no better than a random model).

Model evaluation

Multi-faceted approach: don't rely on a single metric.

Use a suite:

Sensitivity;

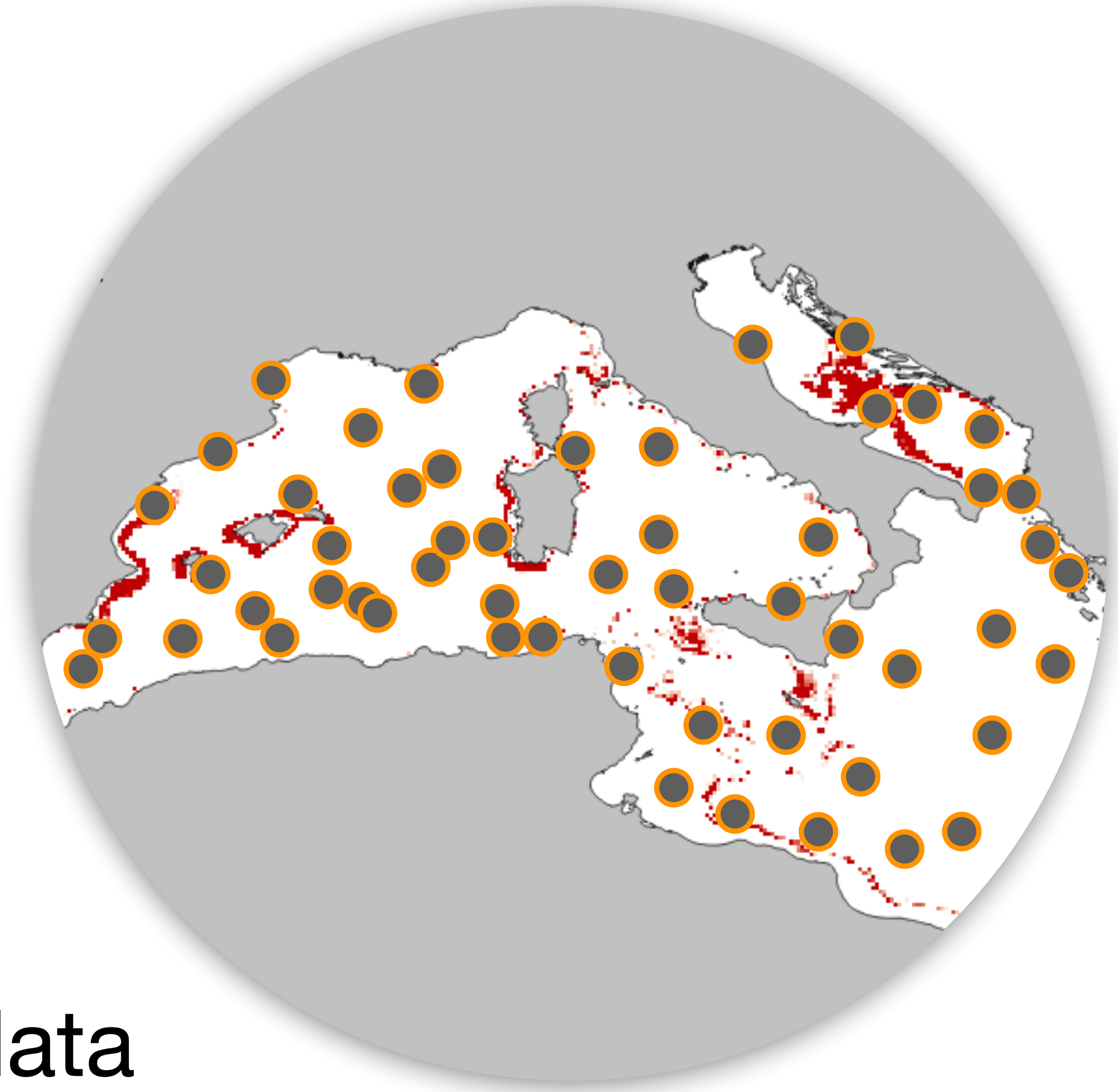
AUC-PRG;

Boyce;

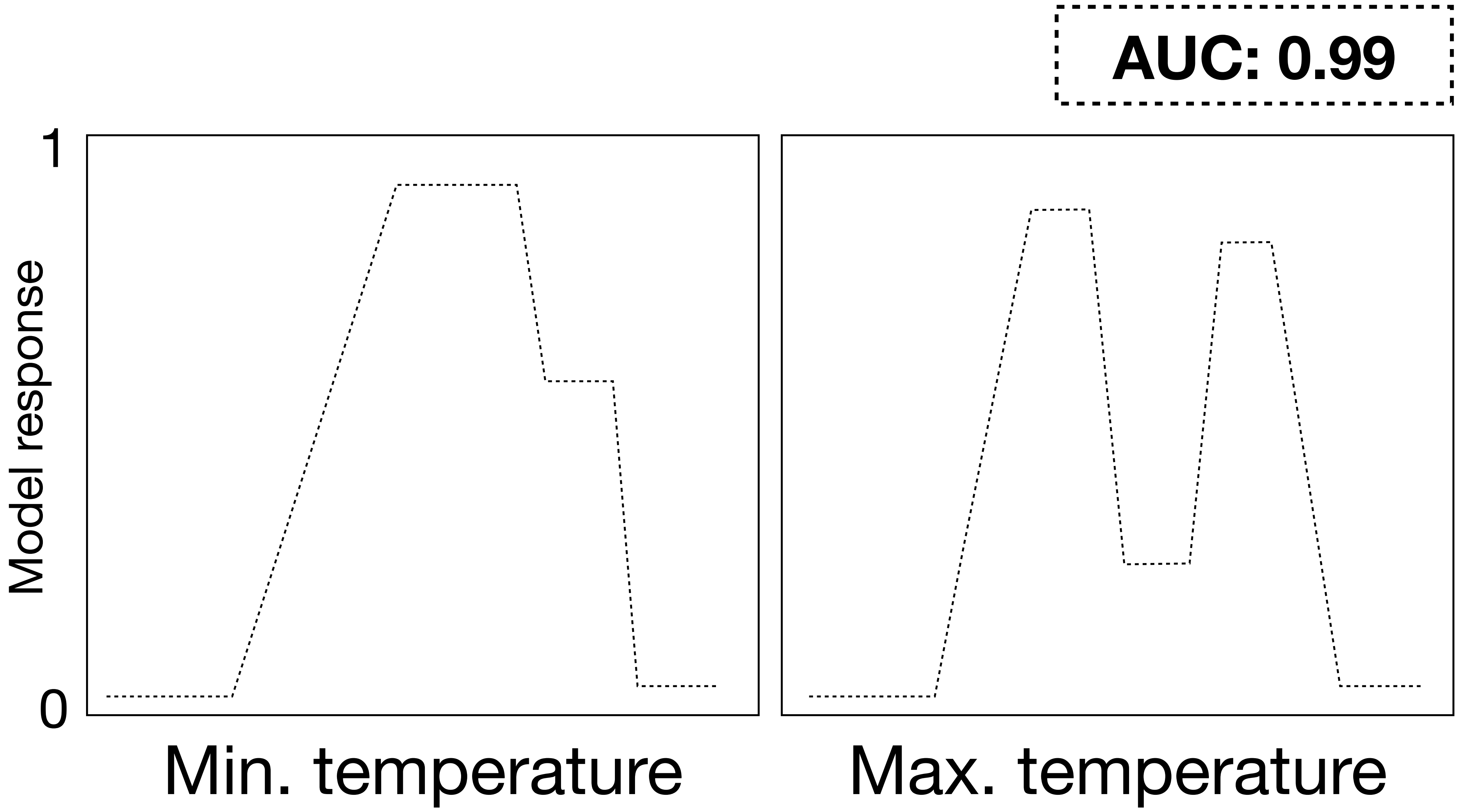
Rigorous Cross-Validation Strategies

High performance linked to good model transferability?

Depends on how it is measured.



● Testing data
○ Training data



Testing accuracy with the training data leads to an overestimation of accuracy, regardless of the model's potential for transferability or if it ecologically sound.
If the models is largely overfitted - the test is performed in the locations of train - no way to assess for generality.

Rigorous Cross-Validation Strategies

High performance linked to good model transferability?

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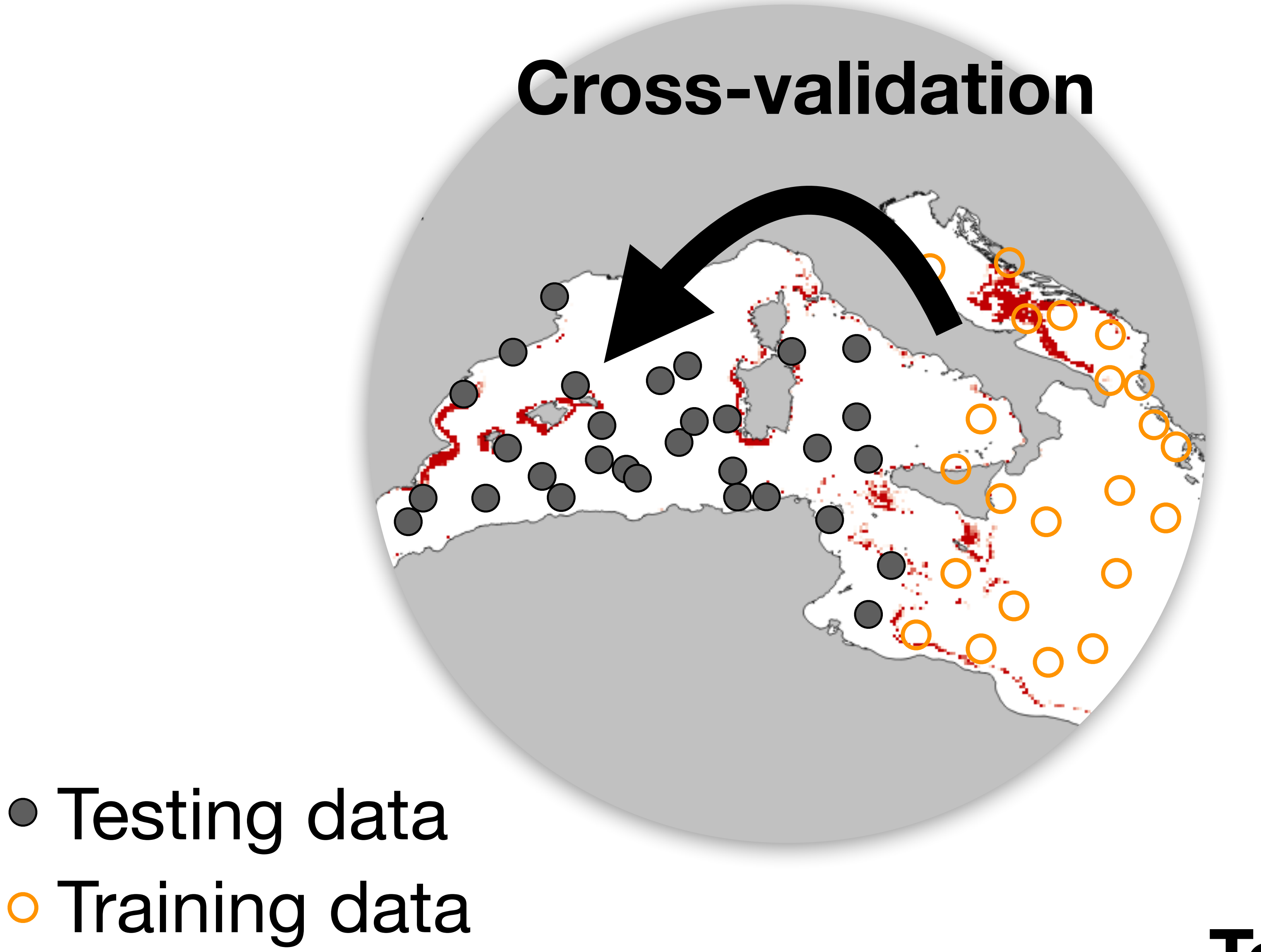
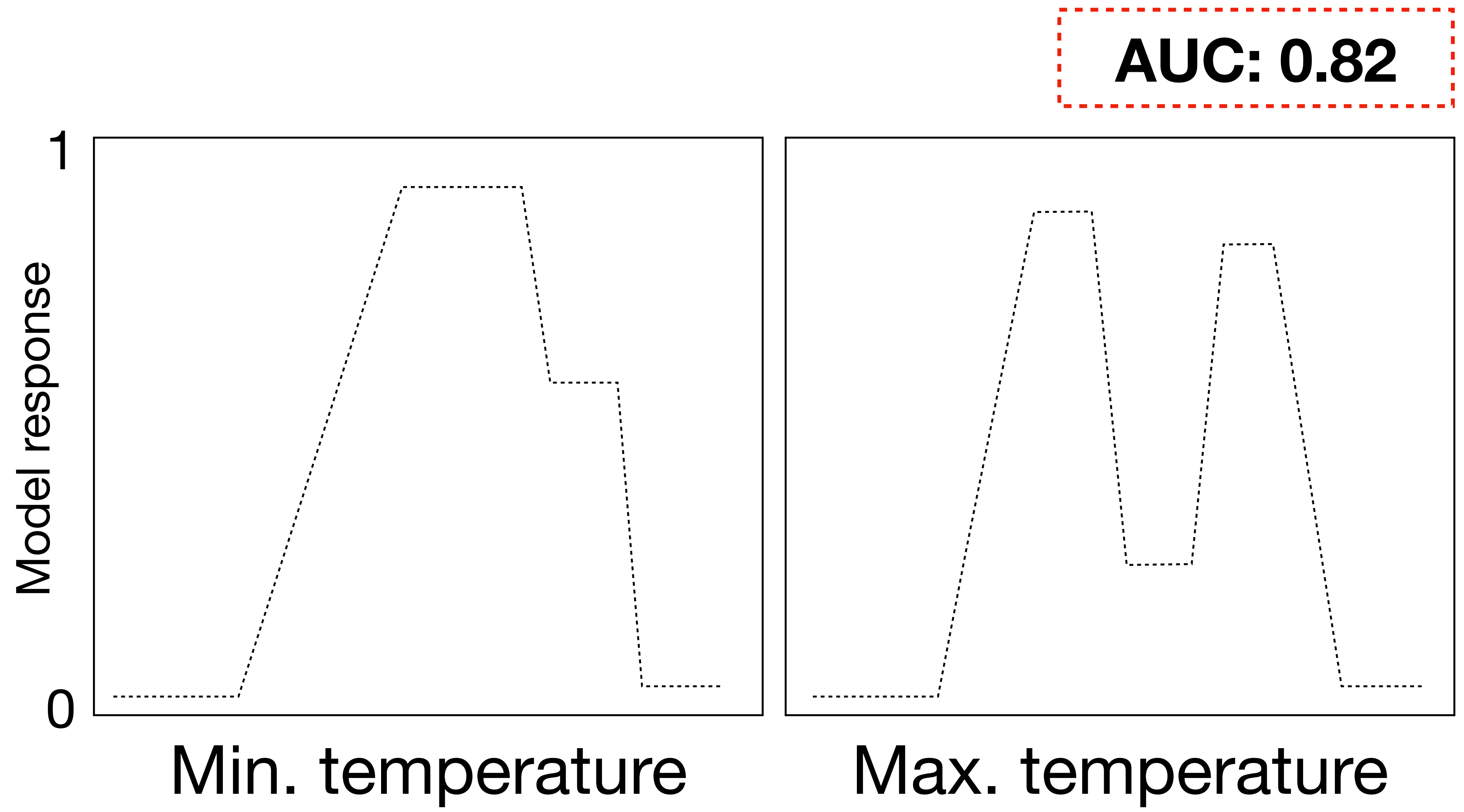




Testing accuracy with independent data is the approach to evaluate if the model generalized the relationship between occurrences and environment, and its transferability.

This is the same as projecting to other places or times.

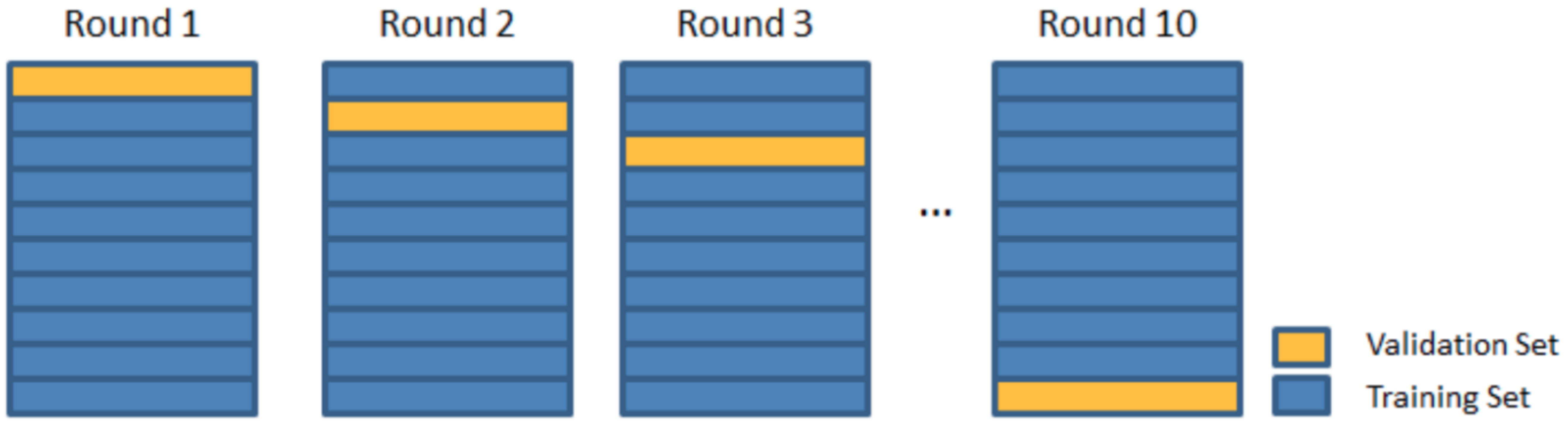
Leads to lower accuracy scores, but more reliable scores.

Missing independent data?

Approach: partitioning the data in k-fold cross-validation interactions, with data splits k times, yielding k estimates of accuracy that can be averaged.

e.g.,

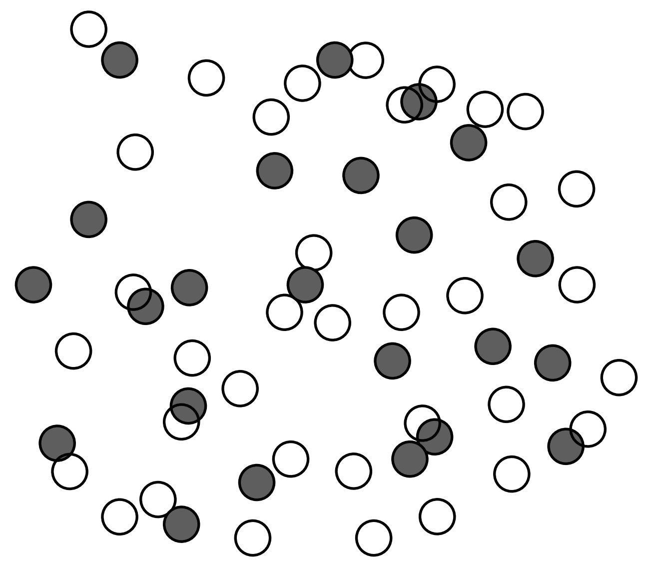
In 10-fold CV, 9/10 of the observations are u**sed to train the model** and the **remaining** 1/10 are **used to estimate performance**; this is repeated ten times and the estimated performance measures are averaged.



Missing independent data

There are different methods to produce independent datasets.

Some approaches provide more independent datasets than others.



Random (70/30)

(70/30 | k-fold)

○ Training data ● Testing data

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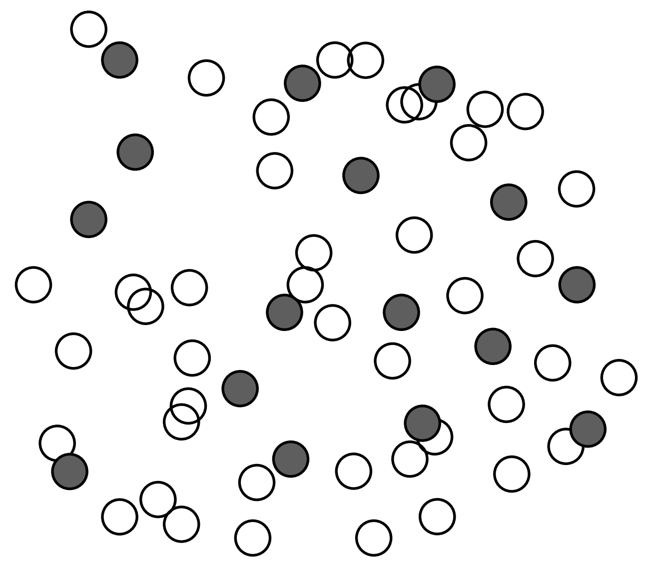

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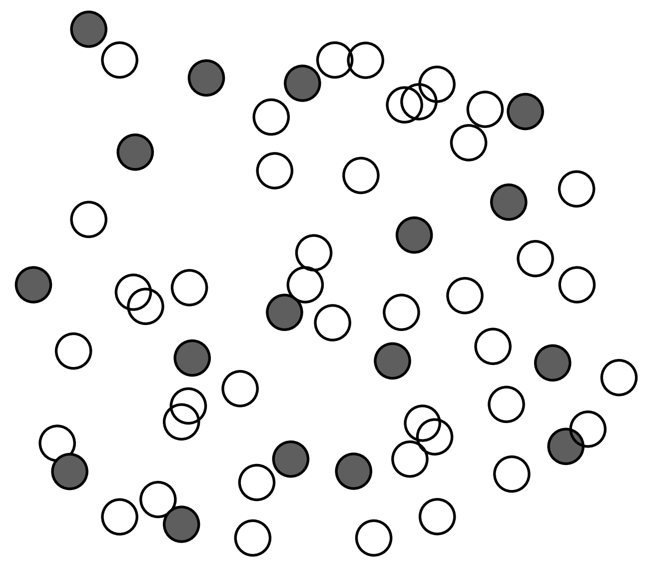


Random (70/30)

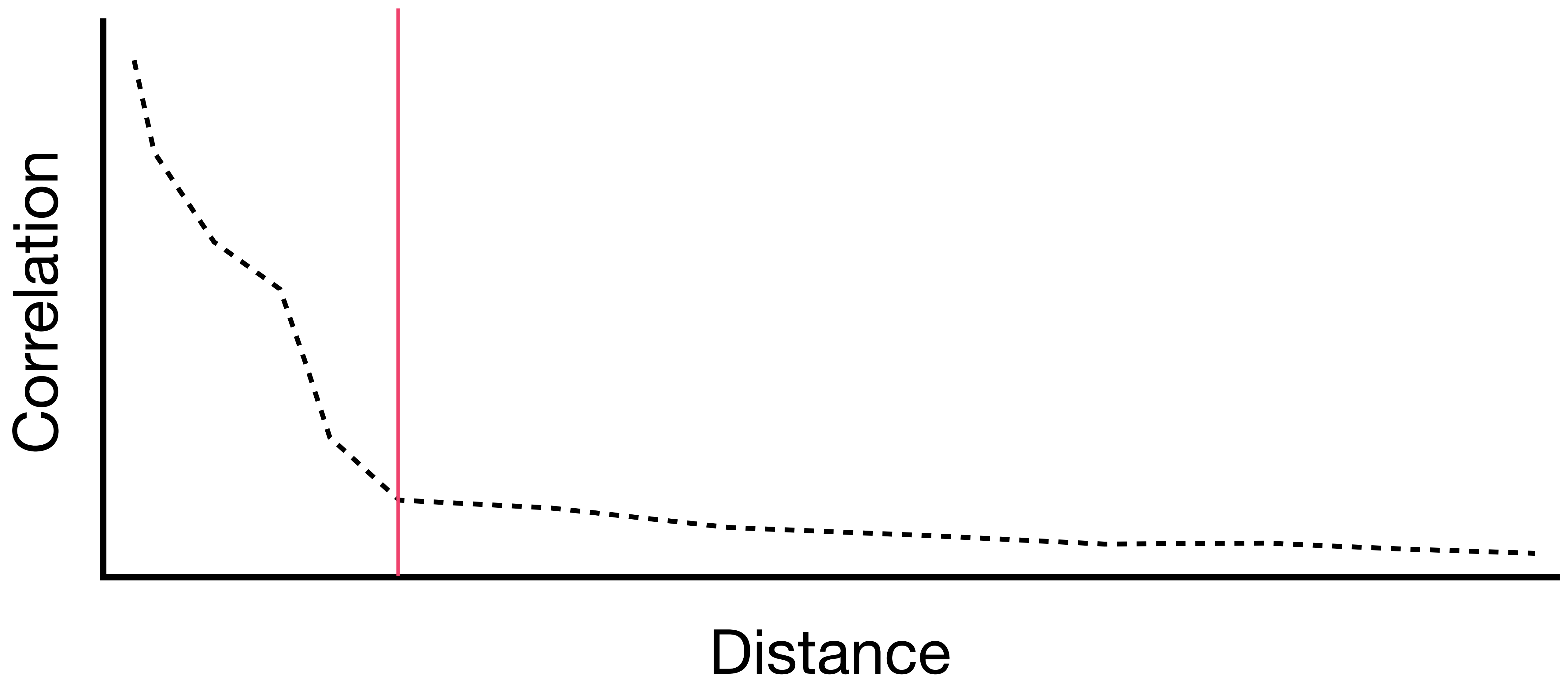
(70/30 | k-fold)

○ Training data ● Testing data

In standard (random) k-fold CV, train and test data are at nearby locations - often not independent.

"*Near things are more related than distant things*".

Can overestimate performance because training and testing folds contain similar data. Violates the independence assumption of statistical tests.



The use of Spatial Autocorrelation (SAC)

Records close to one another share nearly identical environmental conditions.

This clustering inflates model confidence, as the performance assessment accounts for the same environmental datasets multiple time.

We estimate the distance at which records are not SAC - to define independence.

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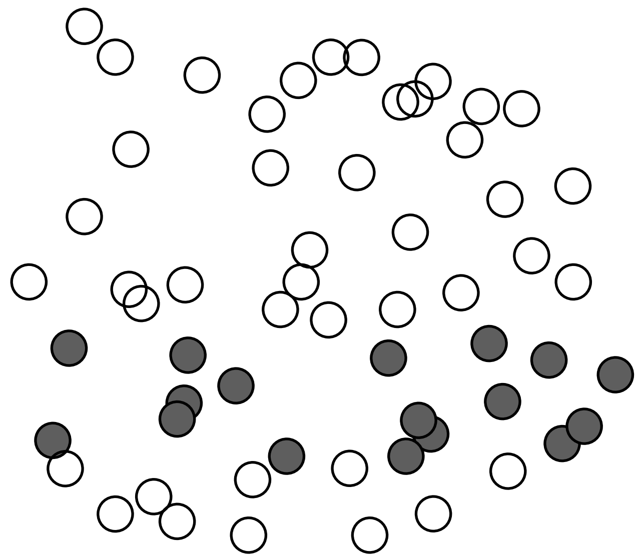
(70/30 | k-fold)



Bands

(latitudinal, longitudinal)

Blocks

(latitudinal, longitudinal)

○ Training data ● Testing data

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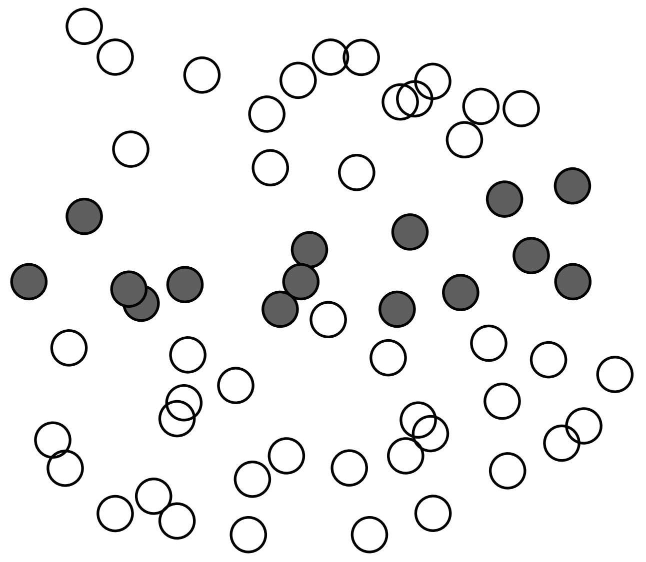

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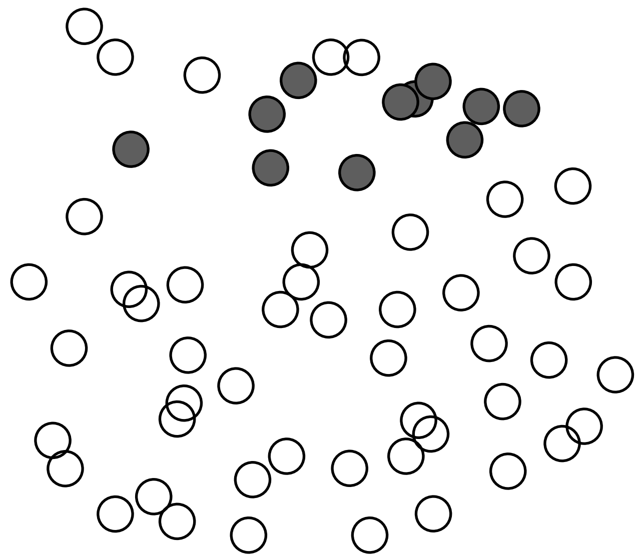

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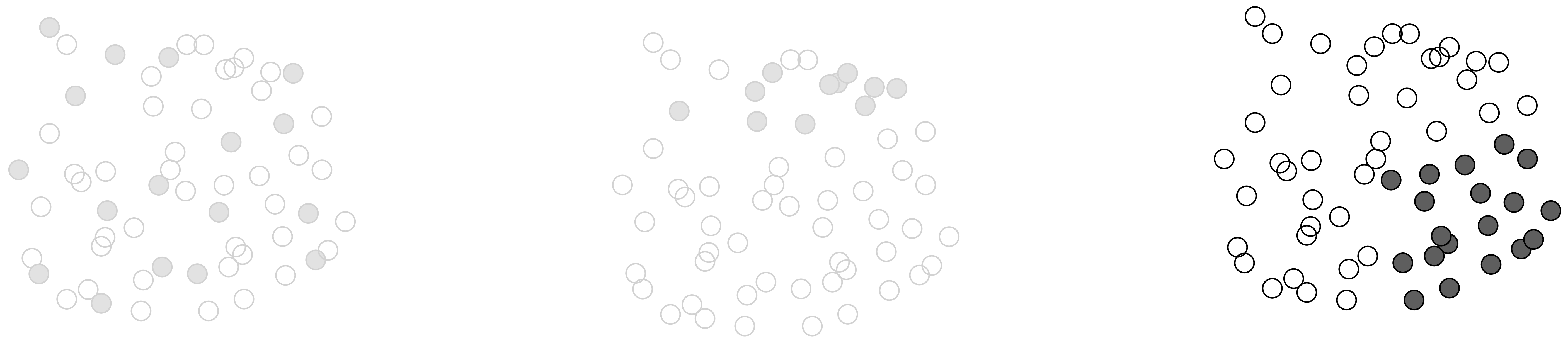


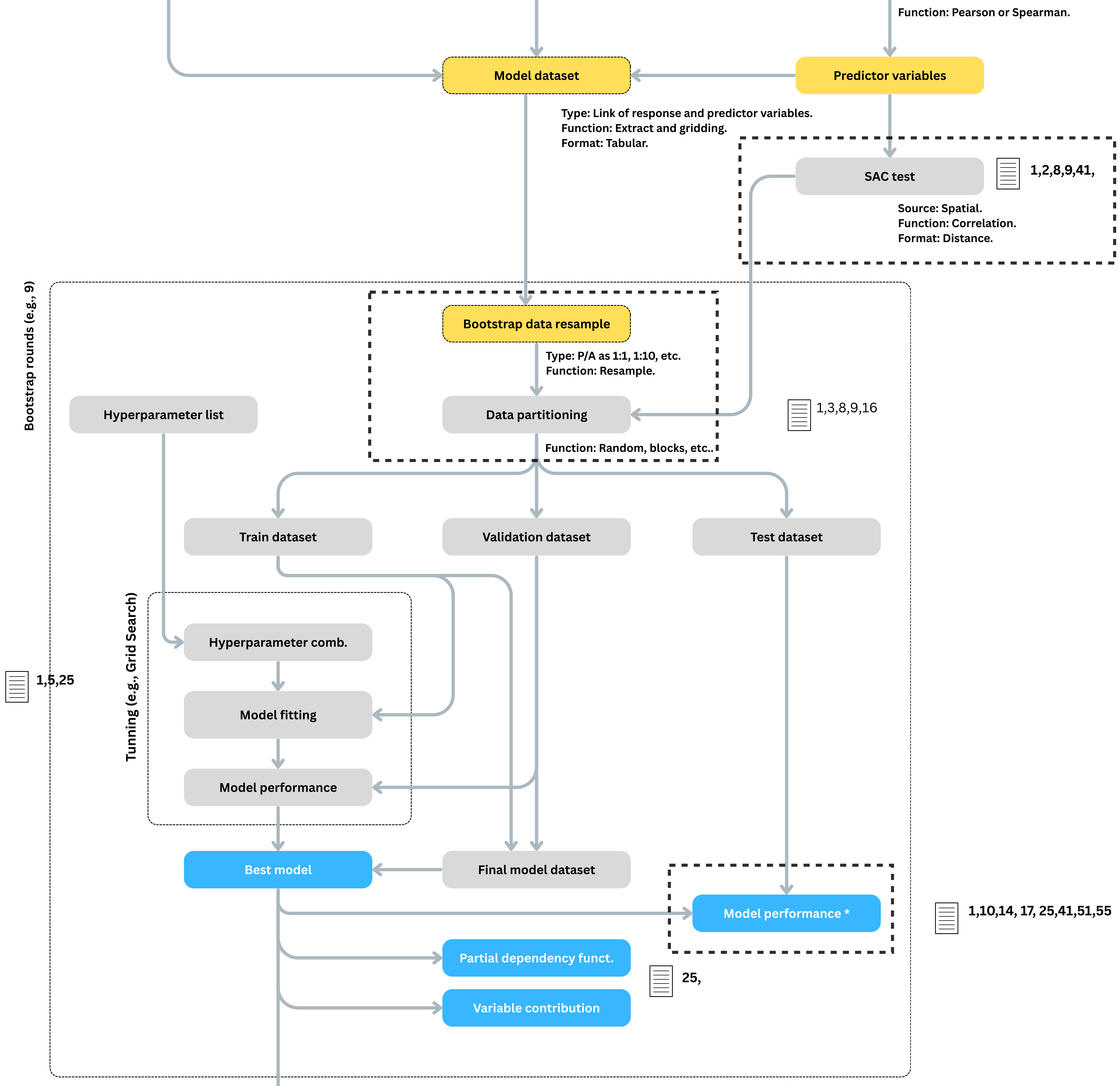
Random (70/30)
(70/30 | k-fold)

Bands
(latitudinal, longitudinal)

Blocks
(latitudinal, longitudinal)

○ Training data ● Testing data

biodiversityDS.



* Model trasferability assessment

Ensuring Ecological Plausibility

Beyond metrics: ecological realism

Does the model make biological sense?

High metric scores (AUC-PR, TSS) are necessary but NOT sufficient!

Ecological plausibility check:

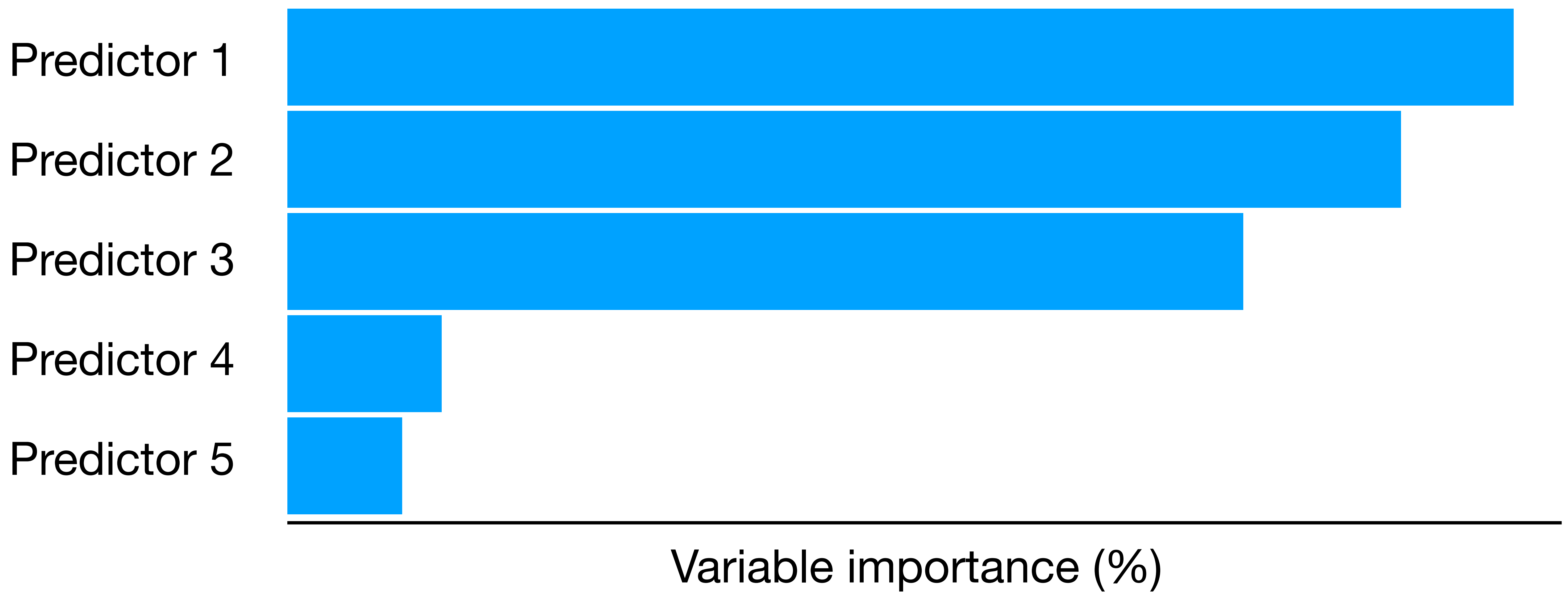
Do predictions align with known species biology, and habitat preferences?

Are the most important predictor variables ecologically relevant?

Relative variable importance

Assists in understanding the contribution of each variable to the model.

The approach is to fit models with and without each variable, in order to determine the potential increase in performance. Without an important variable, a model should reduce its performance.

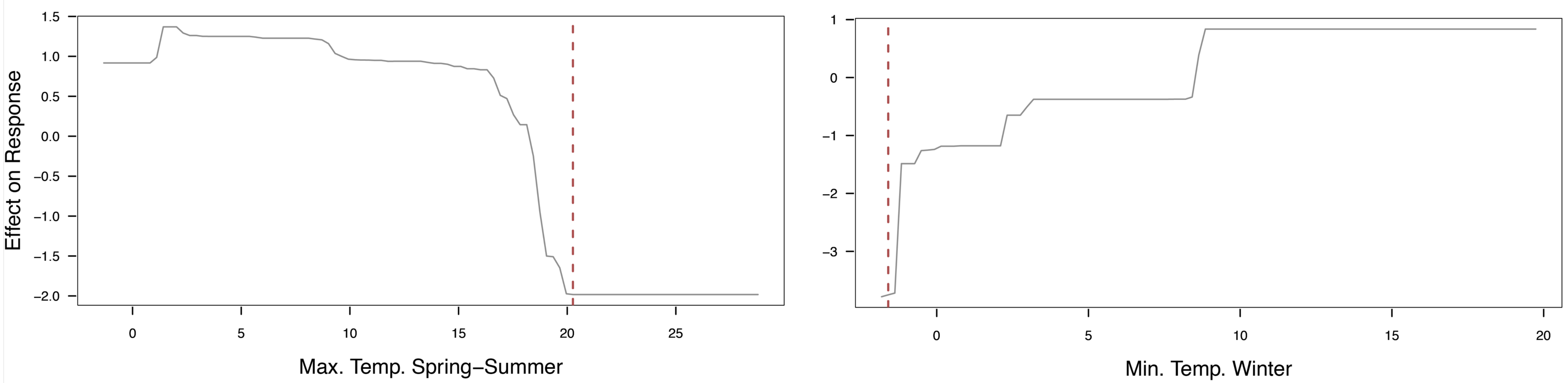


Does the model fit the expectations of ecological theory?

Partial dependency plots

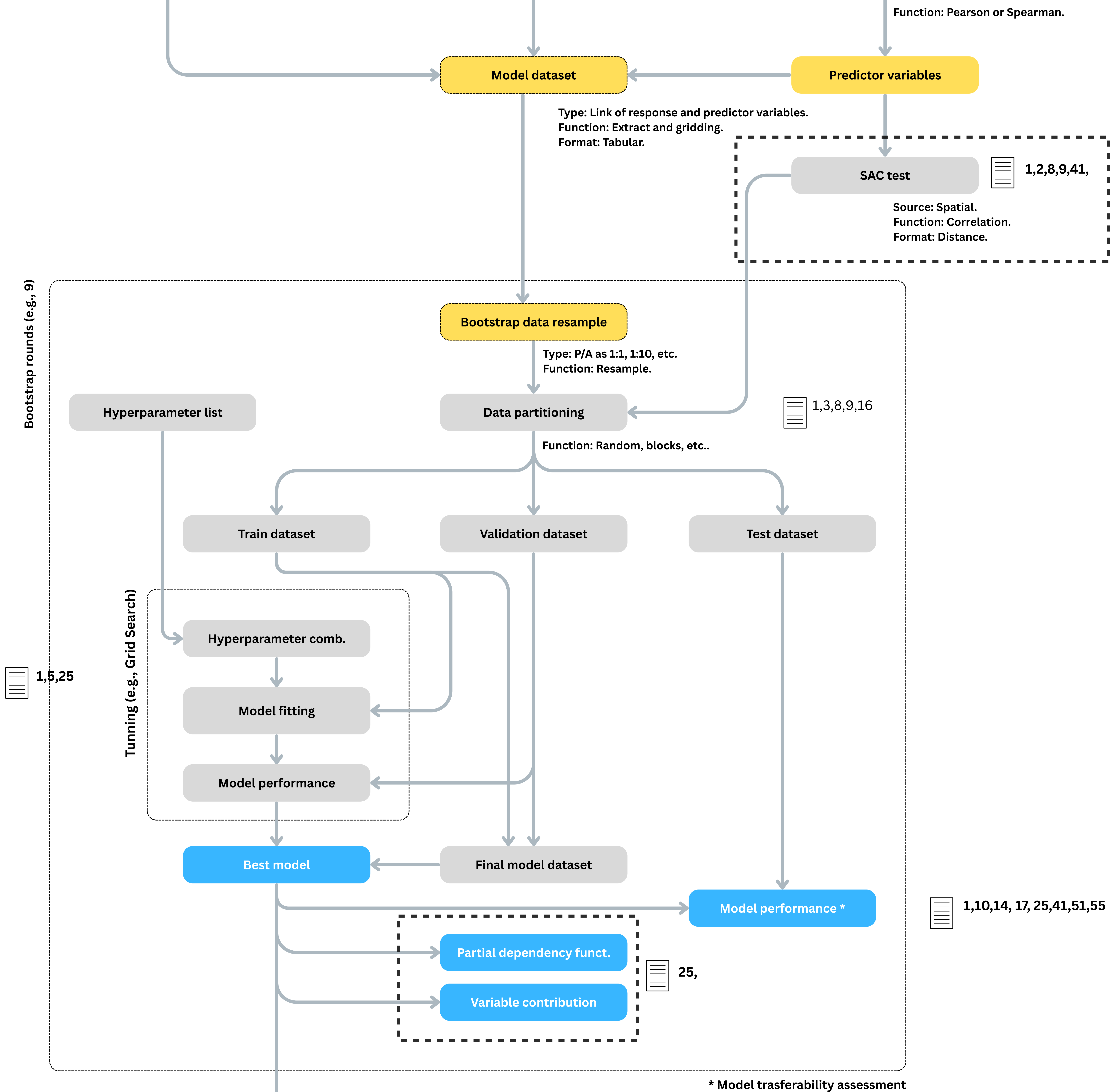
Show the relationship between the probability of occurrence and each variable.

The **response is predicted for one environmental variable while the other environmental variables are held constant at their mean.** The x-axis represents the range of values of the environmental variable, and the y-axis gives the probability of occurrence on a scale from 0 (low probability) to 1 (high probability).



Does the model fit the expectations of ecological theory?

Curve and limiting points match physiological data (reliable model!)



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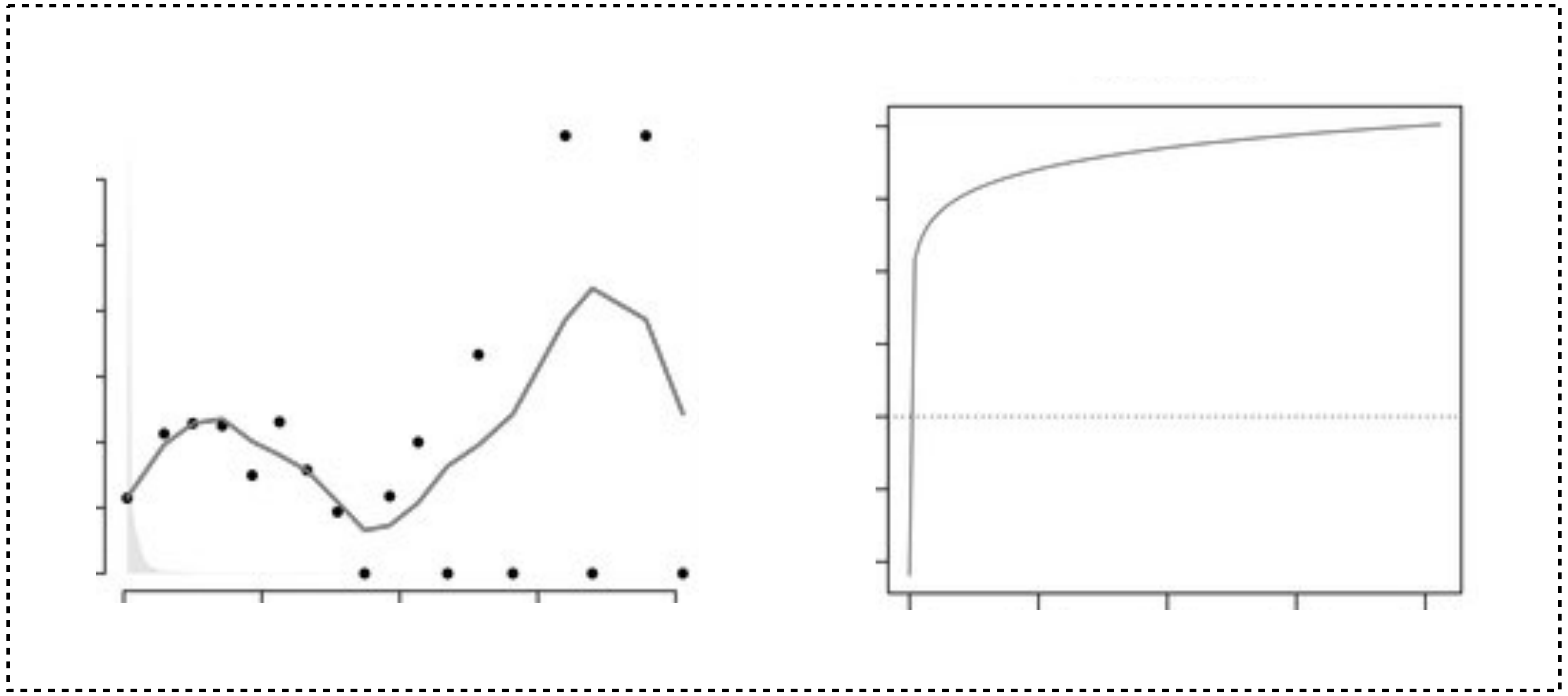
Overfitted response vs. positive monotonicity



Enforcing Ecological Rules (Monotonic Constraints)

A way to force the modeled relationship to follow a pre-defined, simple trend.

Prevents the model from fitting overly complex, potentially unrealistic shapes to the data for a particular variable.

The choice of monotonicity constraints is made a priori per variable **based on ecological theory** (no monotonicity, negative or positive monotonicity).

Only a few machine learning algorithms have such implementation (BRT; XGBoost).

Model Transferability and Environmental Novelty

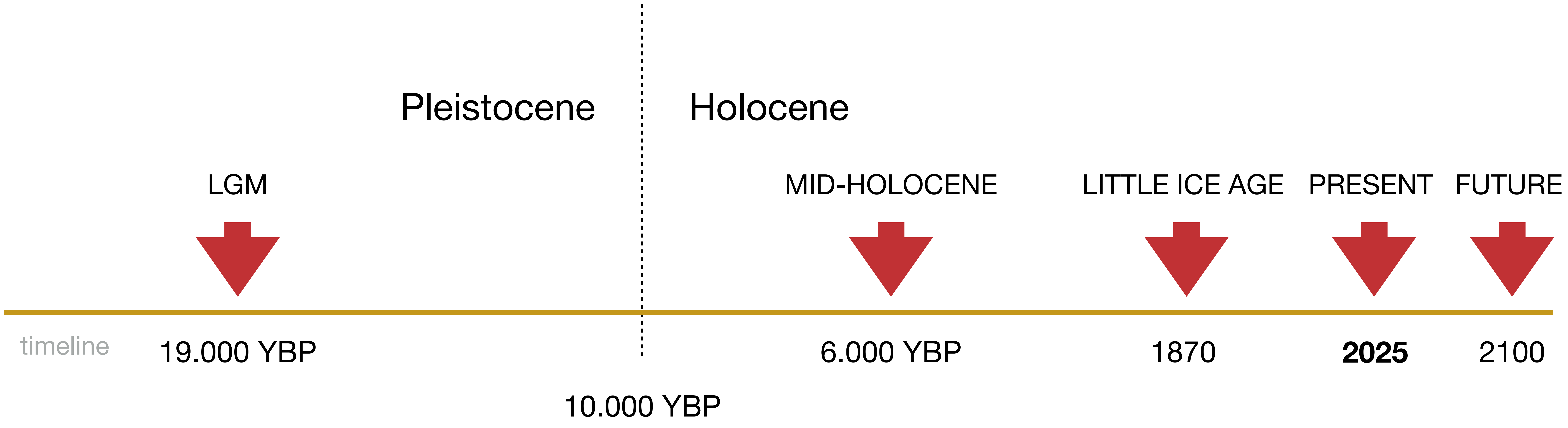
Model transferability

Ability of a model to predict beyond the data it was originally trained on.

Particularly **relevant when forecasting species distributions** under **climate change**.

Mainly dependent on:

1. Adequacy of environmental and species occurrence data to train the models;
2. Reduced complexity of the models (e.g., number of variables);
2. **Proper choice / tuning of model parameterization.**



Predictive model-based transferability

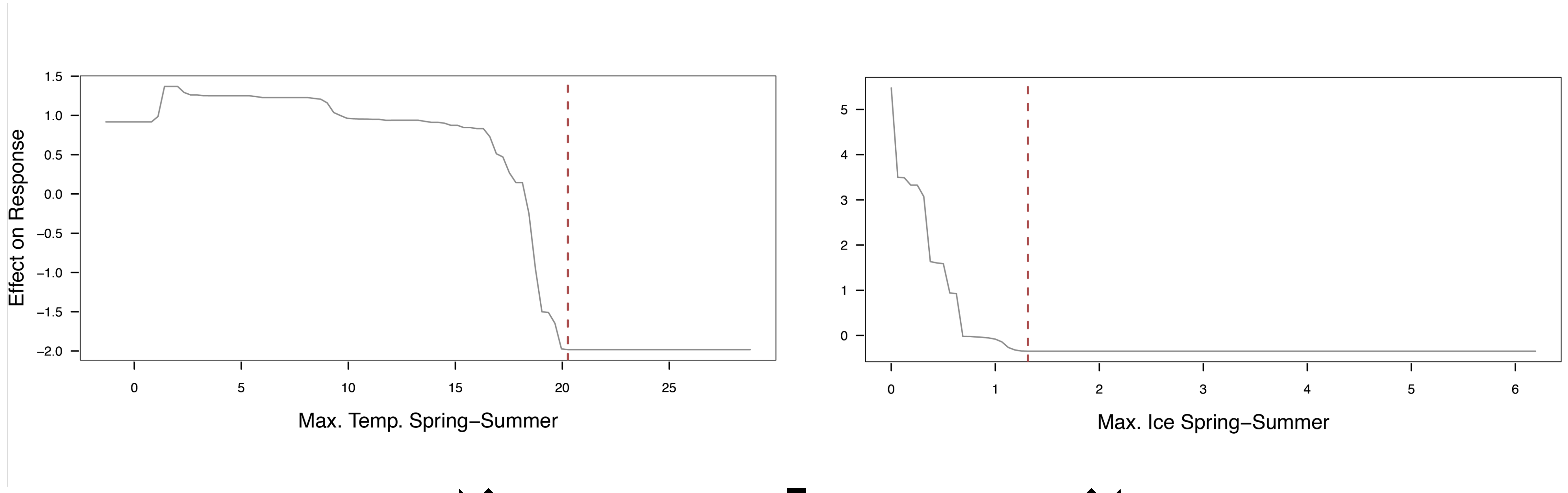
Ability of a model to predict beyond the data it was originally trained on.

Process: a model fits **observed species occurrence data against environmental variables for the corresponding period**. The **same model is then applied directly to new climate data** representing new environmental conditions.

Assis et al. 2024

Predictive model-based transferability

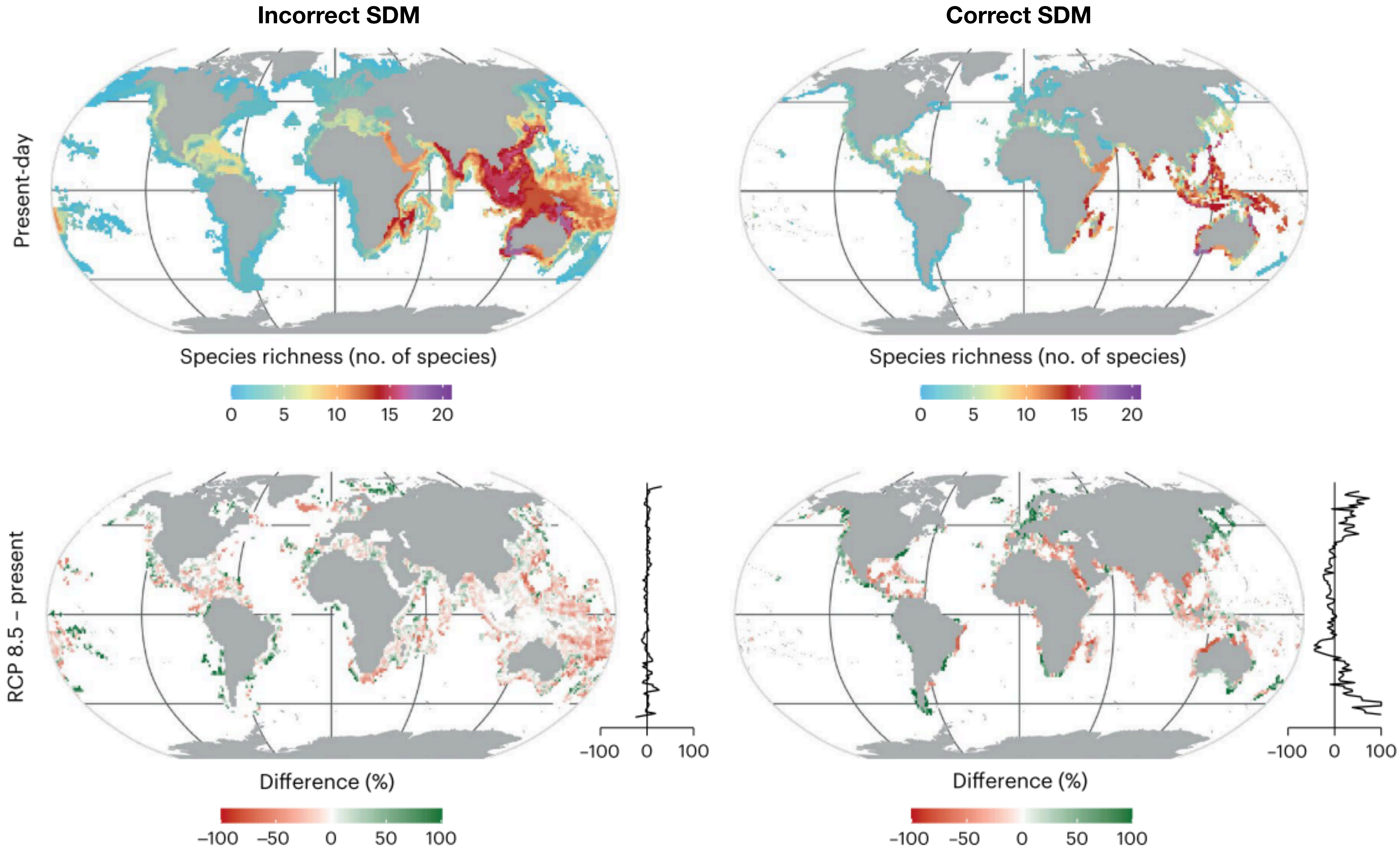
Fitted functions



Layers past	Layers present	Layers future
Past distribution	Present distribution	Future distribution

Regardless of the time period / region, **all environmental layers must be included in the transferability process**. So, the **availability of layers for climate scenarios also determines the choice of predictors for model fitting**.

Assis et al. 2024

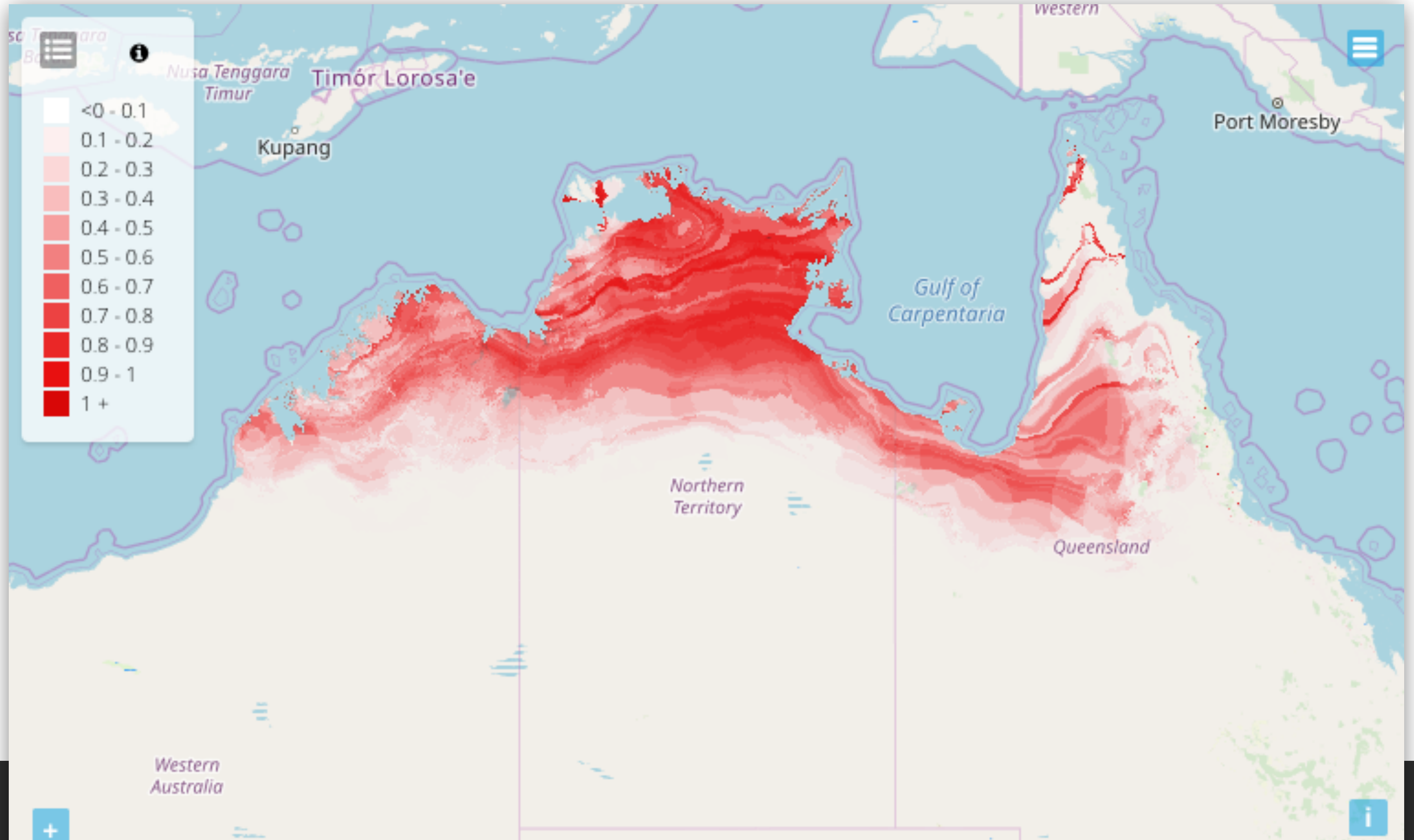


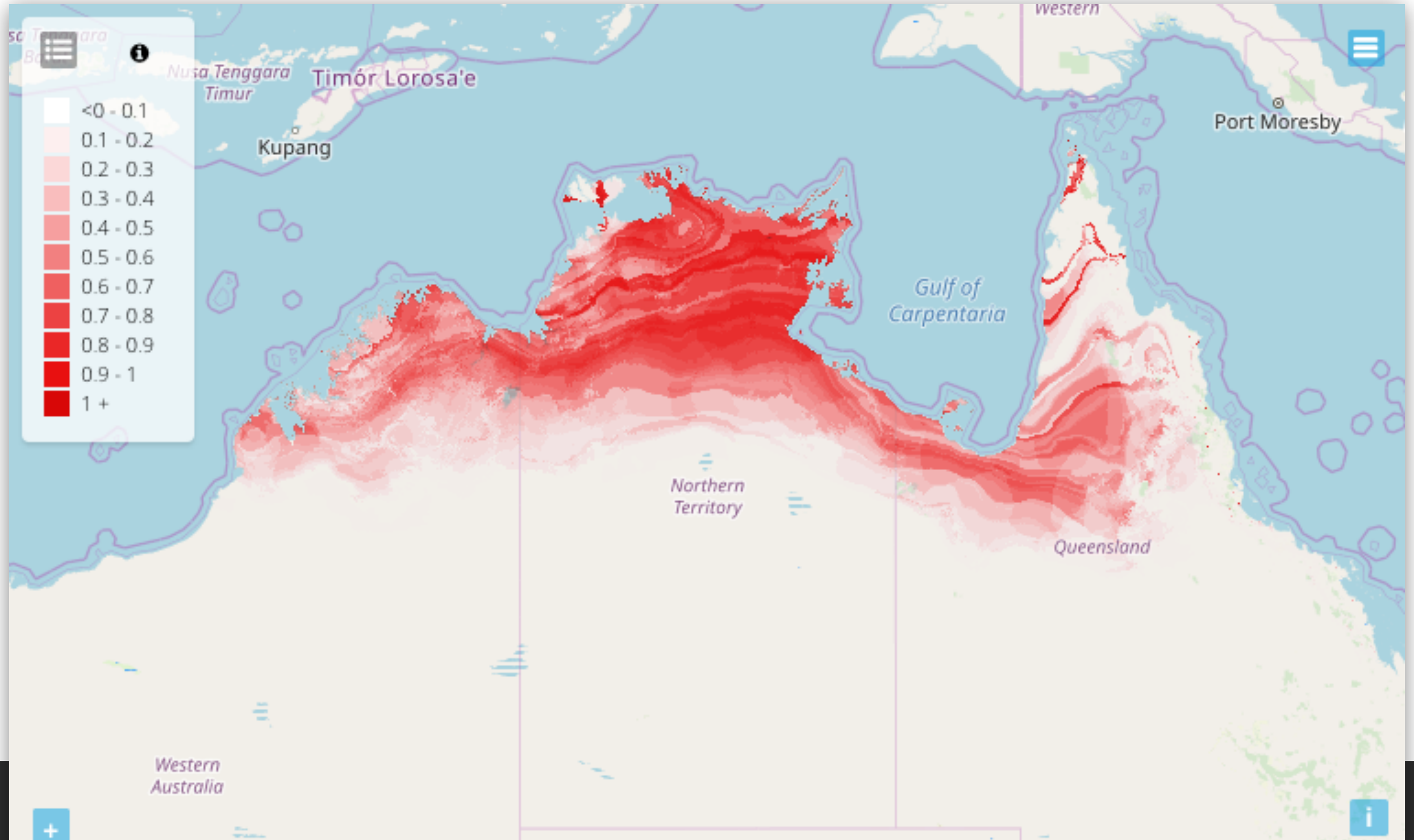
Assis et al. 2024

Predictive model-based transferability

Model transferability can be analyzed (per cell) with different approaches:

(1) comparing **predictions** (e.g., probability) under different conditions.



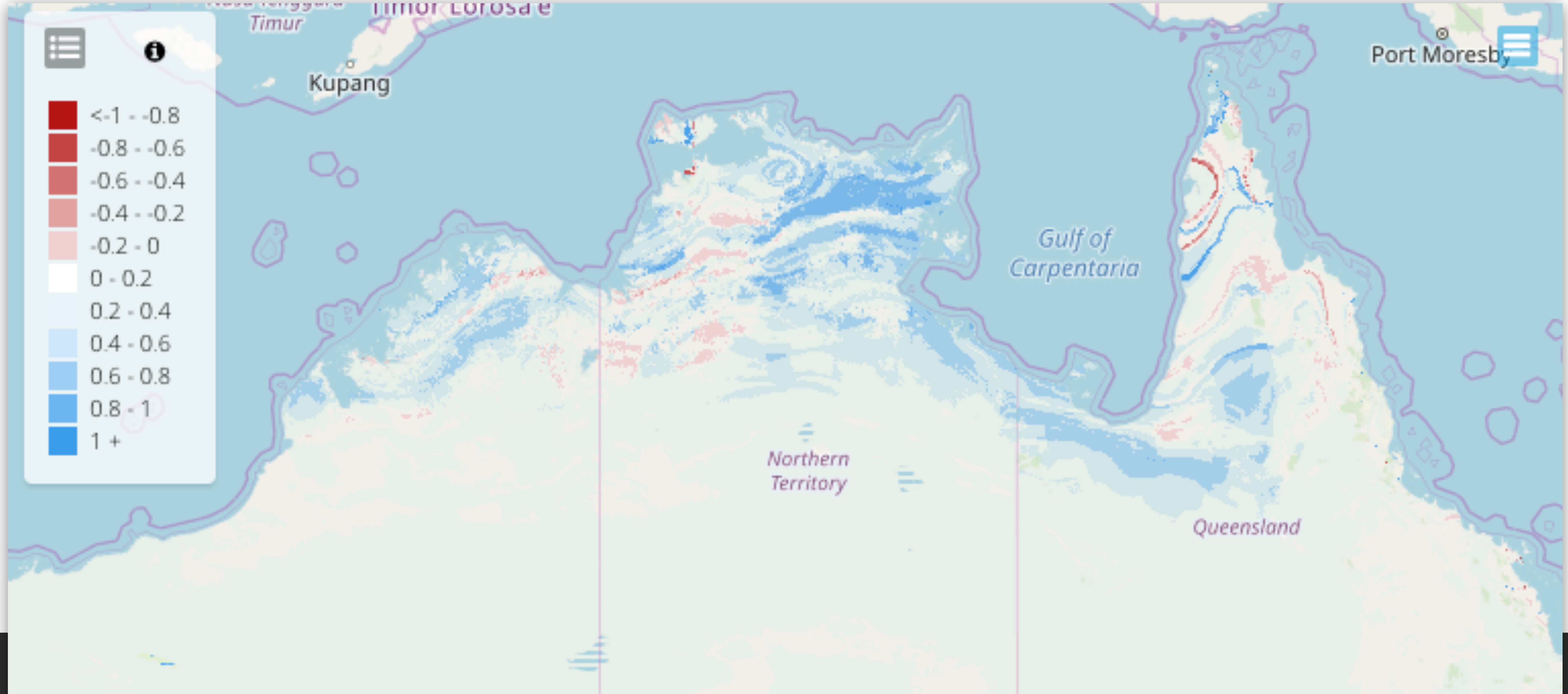


(2) **change in probability**, i.e., the difference in the predicted probabilities of different conditions;

e.g.,

[Layer 1 Future probability] **-** [Layer 2 present-day probability]

The map scales from -1 to 1, where negative values refer to lower suitable conditions and positive higher suitability.



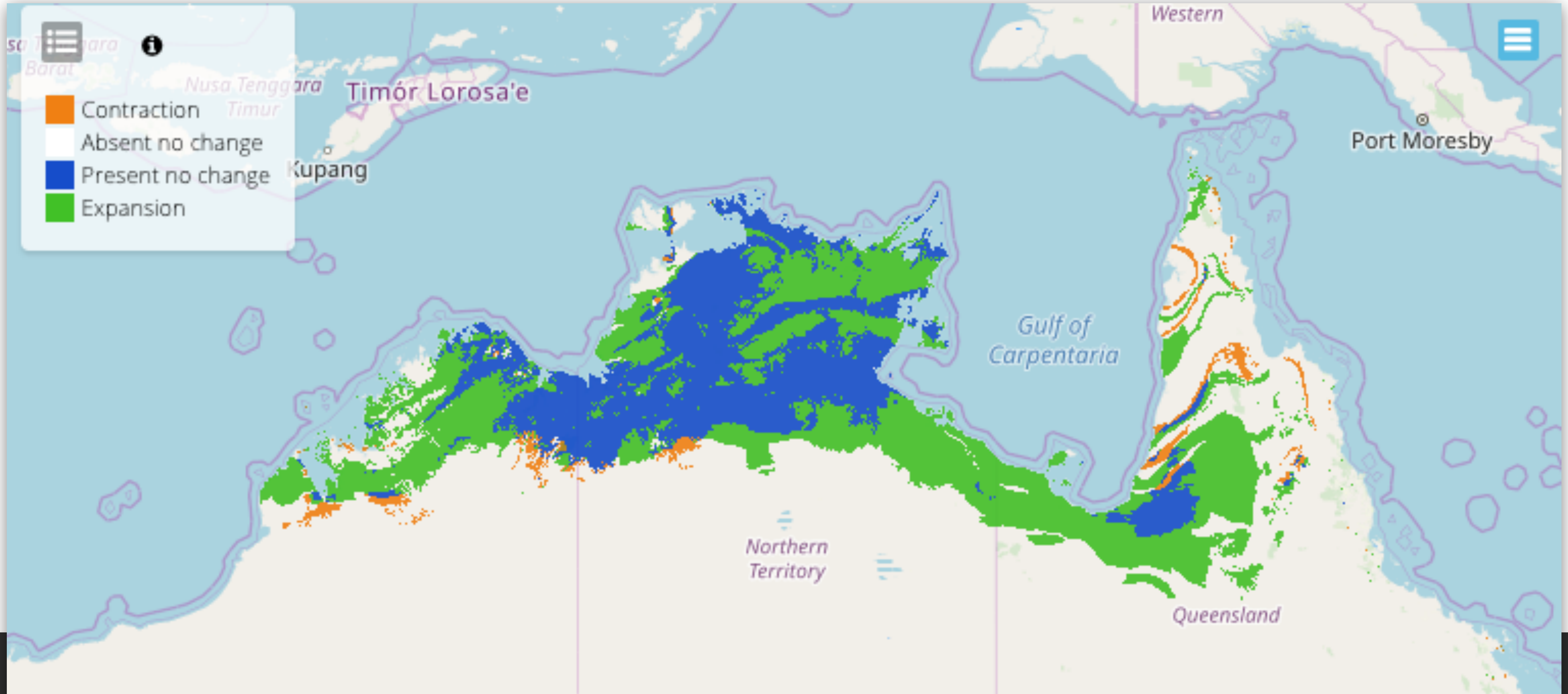
(3) **change in species distributional range**, generated with binary maps (i.e., 0 or 1).

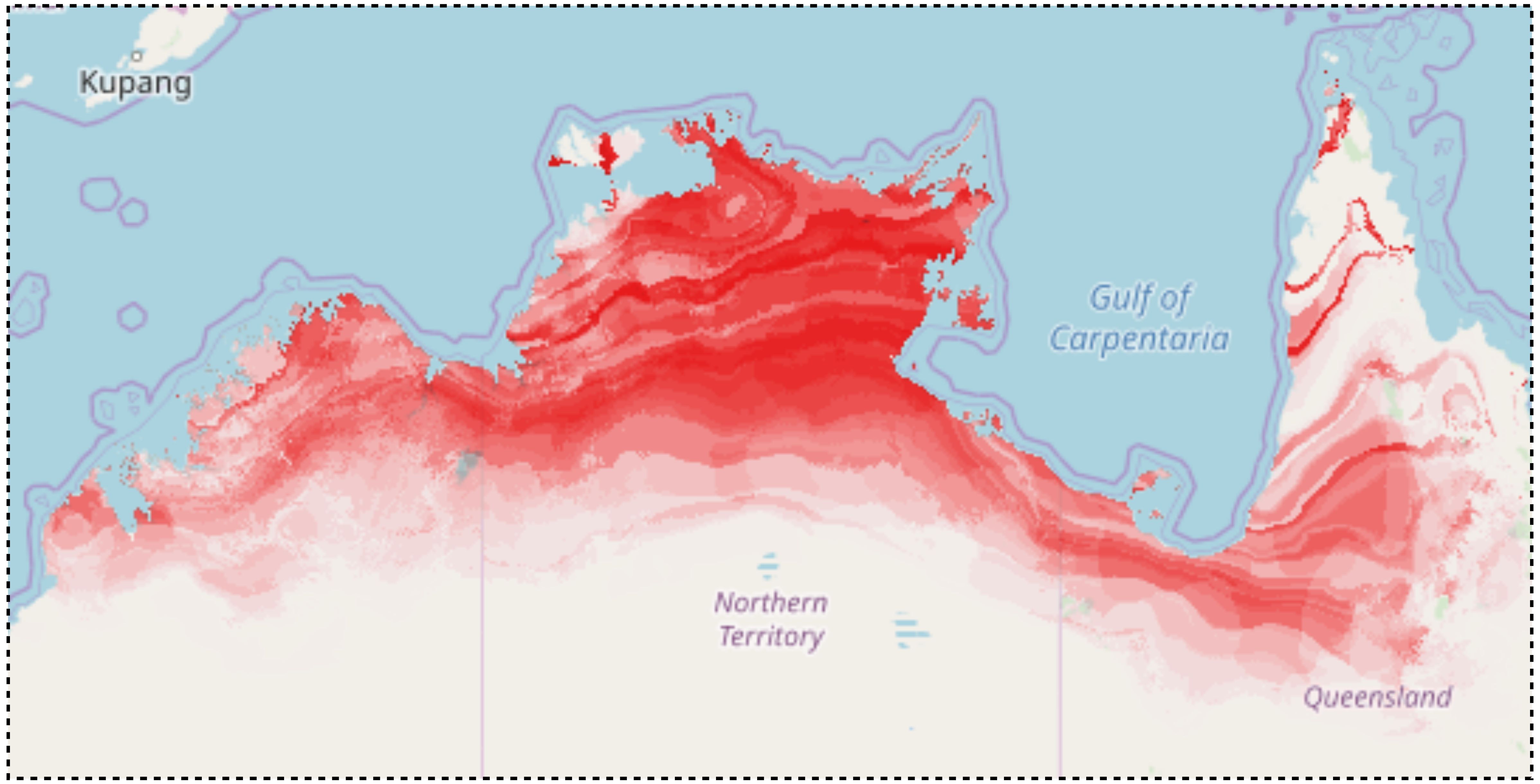
Comparing maps can indicate:

No change: present distribution equals future distribution;

Range loss or contraction: presence in the present and absence in future;

Range expansion: absence in the present and presence in the future.



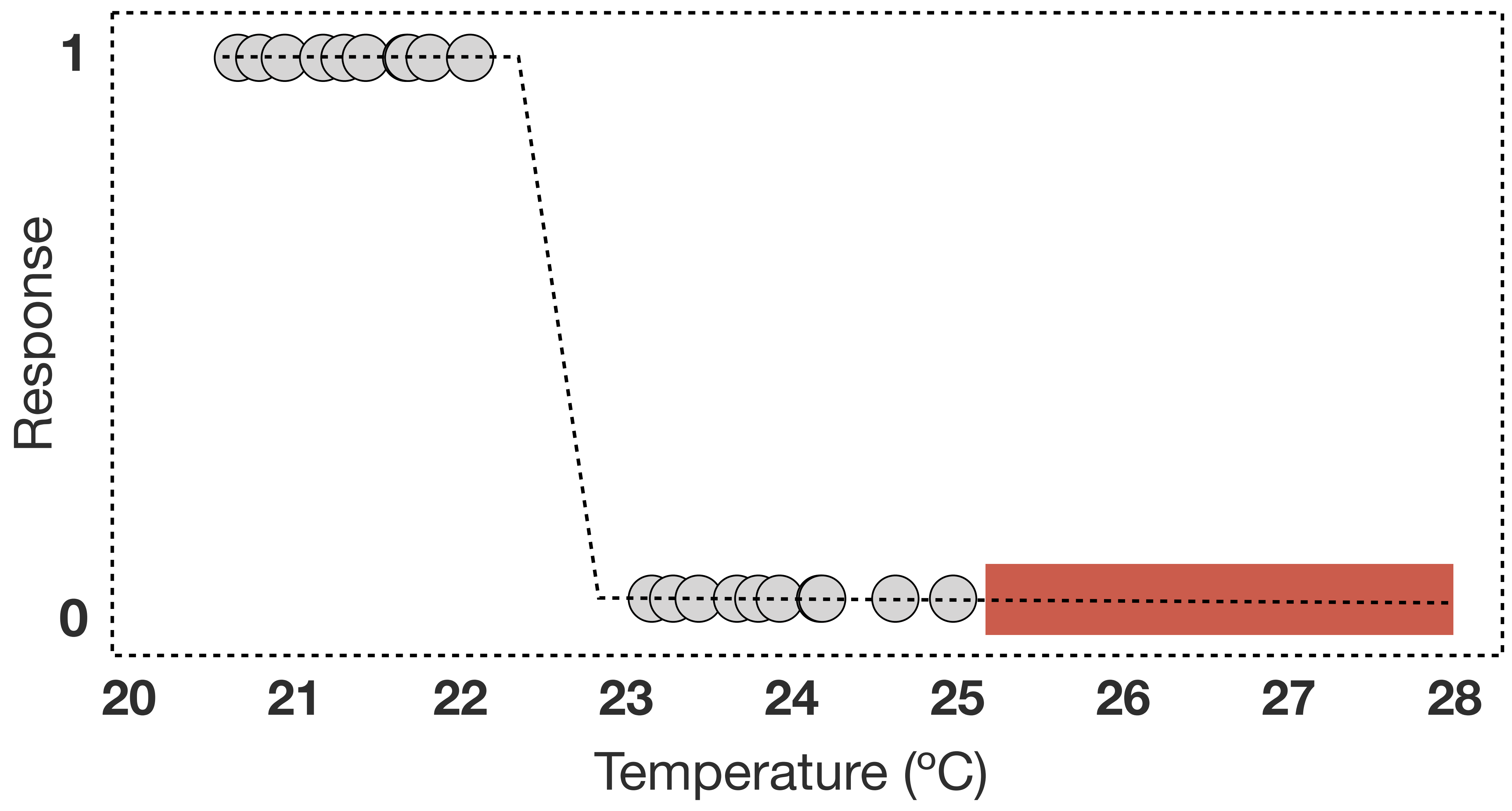


Predictive model-based interpolation

Predictions to **new sites within the range of values of environmental conditions sampled in the training data**.

From **scattered presence records to continuous distribution surfaces**.

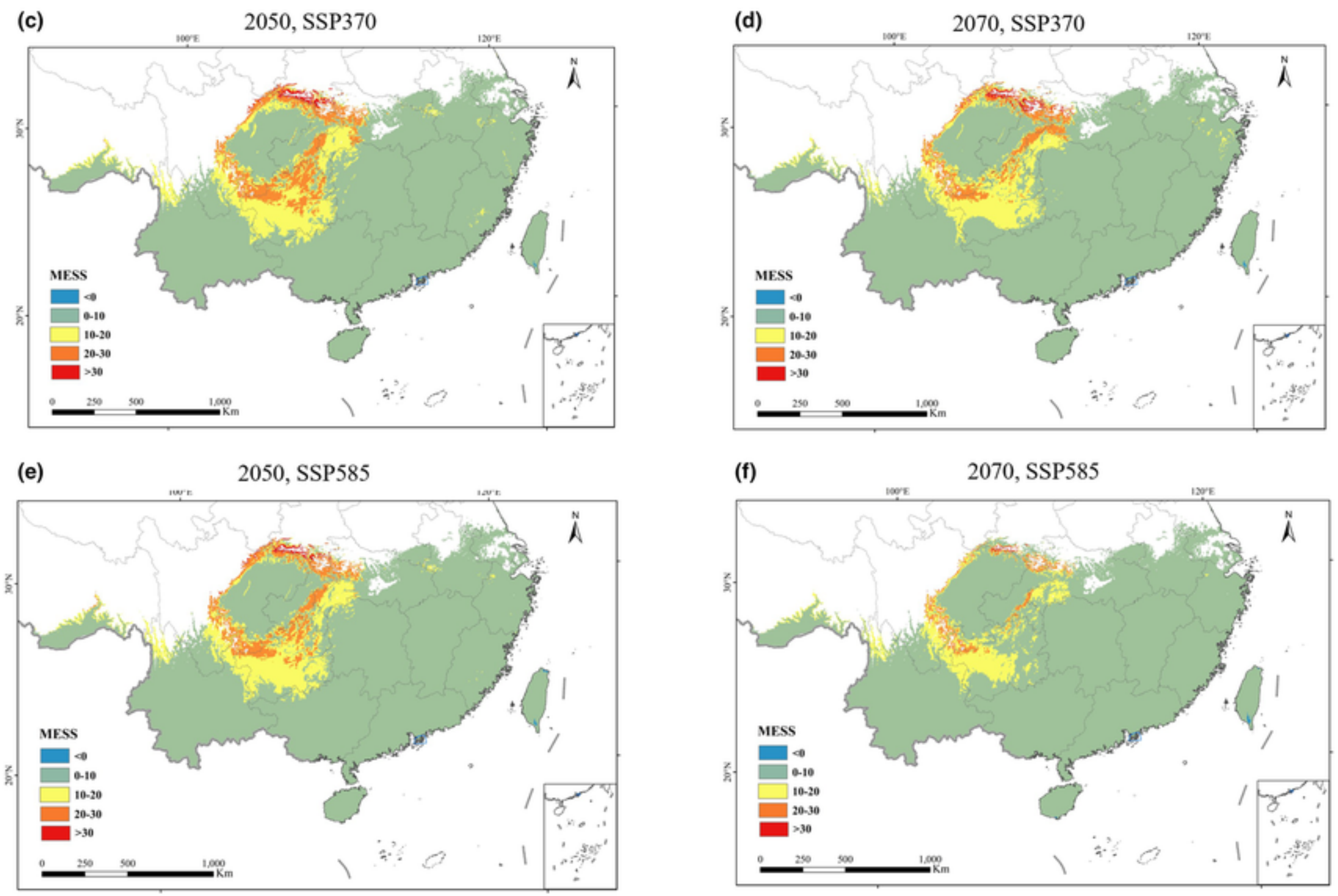
Maps do not show where species occur, but rather the distribution of regions with suitable environmental conditions.



Model transferability may lead to extrapolation, i.e., predictions to environmental values outside the range of environmental values used to fit the model.

e.g.,

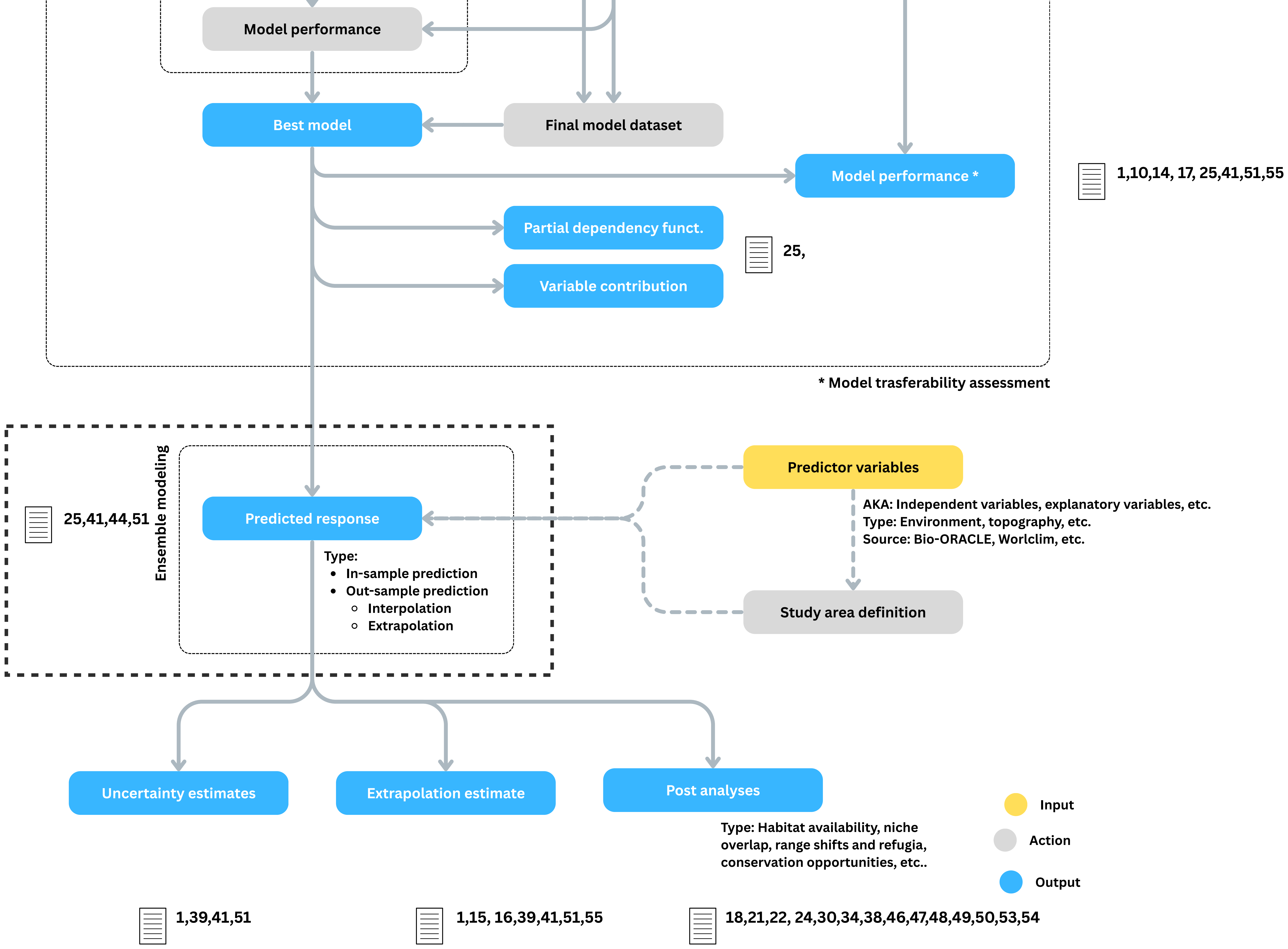
A **model fitted records with temperatures of 20-25ºC**. If **predictions** are **made for temperatures > 25ºC**, then the model will extrapolate. No information exists on the probability of occurrence at > 25ºC, so the **predictions will be uncertain**.



Quantifying Environmental Novelty

Future climates or new regions often contain conditions the model has never seen.

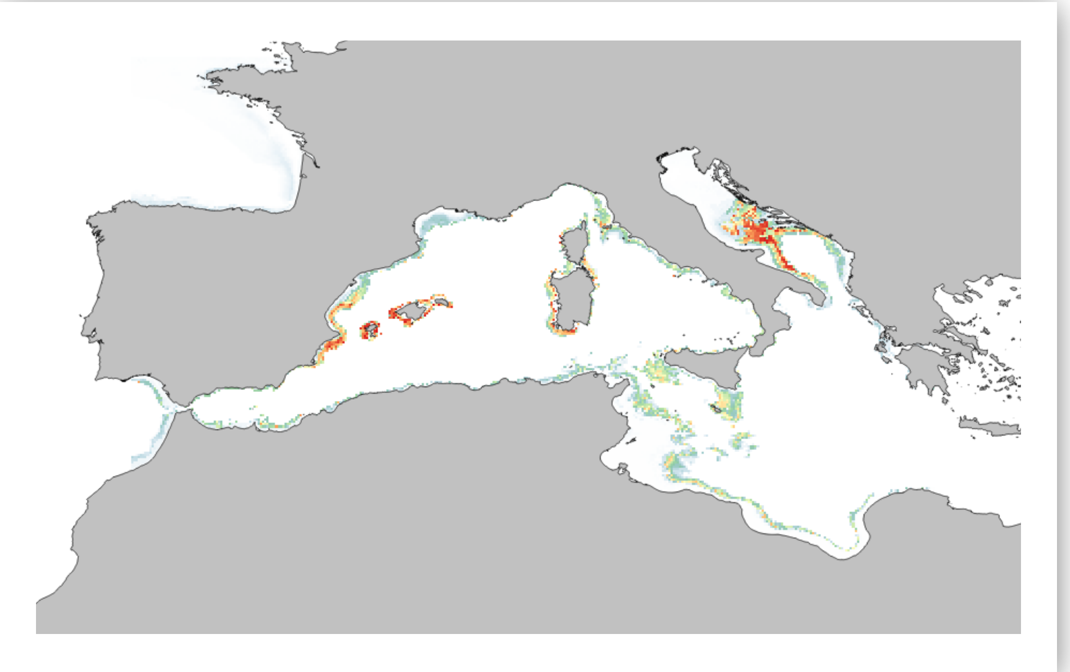
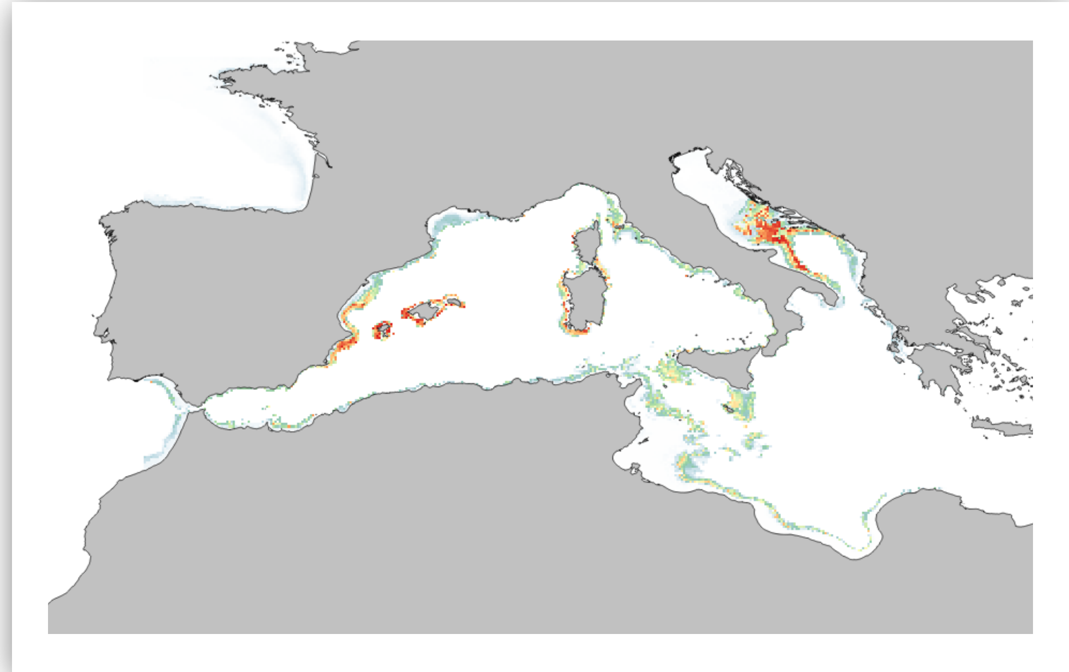
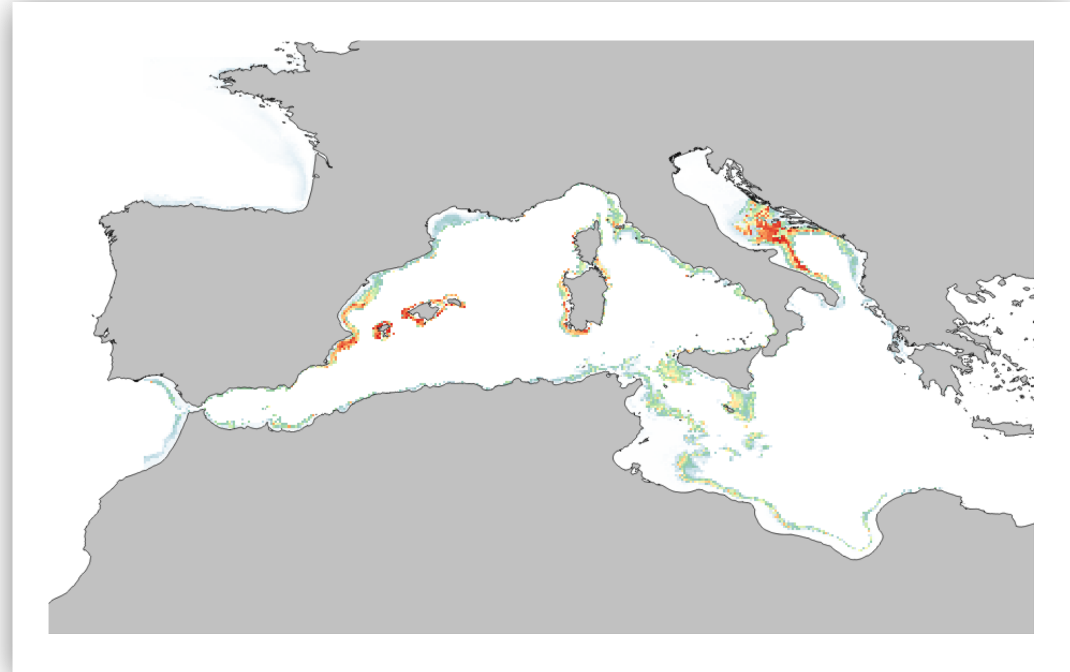
Multivariate Environmental Similarity Surfaces: A technique to map how environmentally similar (or dissimilar) the projection area is compared to the calibration area.


Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
25,
Variable contribution
* Model trasferability assessment
Ensemble modeling
25,41,44,51
Predicted response
Type:
In-sample prediction
Out-sample prediction
Interpolation
Extrapolation
Predictor variables
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Study area definition
Uncertainty estimates
Extrapolation estimate
Post analyses
Type: Habitat availability, niche overlap, range shifts and refugia, conservation opportunities, etc..
Input
Action
Output
1,39,41,51
1,15, 16,39,41,51,55
18,21,22, 24,30,34,38,46,47,48,49,50,53,54

Uncertainty estimates and Ensemble Modeling

Ensembling models

Instead of using a single-model to investigate species distributions, one can conduct an **ensemble experiment**. This **reduces the uncertainty of algorithms by combining their outputs with a general statistics** (e.g., mean).

mean( **,**  **) =** 

Mapping uncertainty between models

Ensembles allow us to map where different algorithms agree and where they heavily disagree. By mapping the standard deviation between the models, we can visually identify high-confidence habitats versus highly uncertain regions

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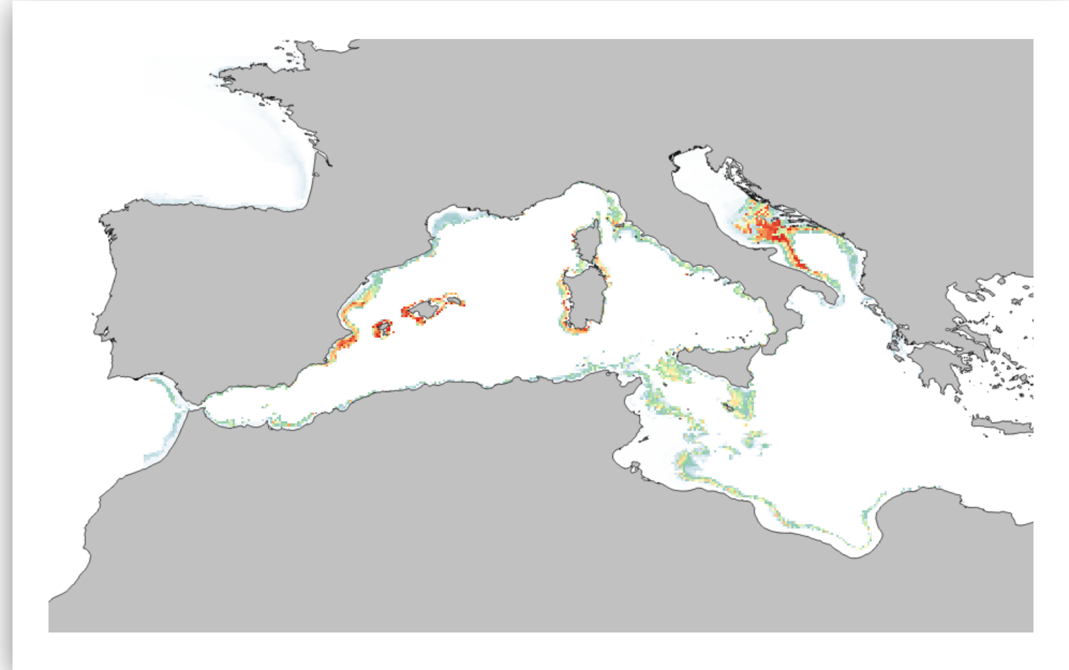

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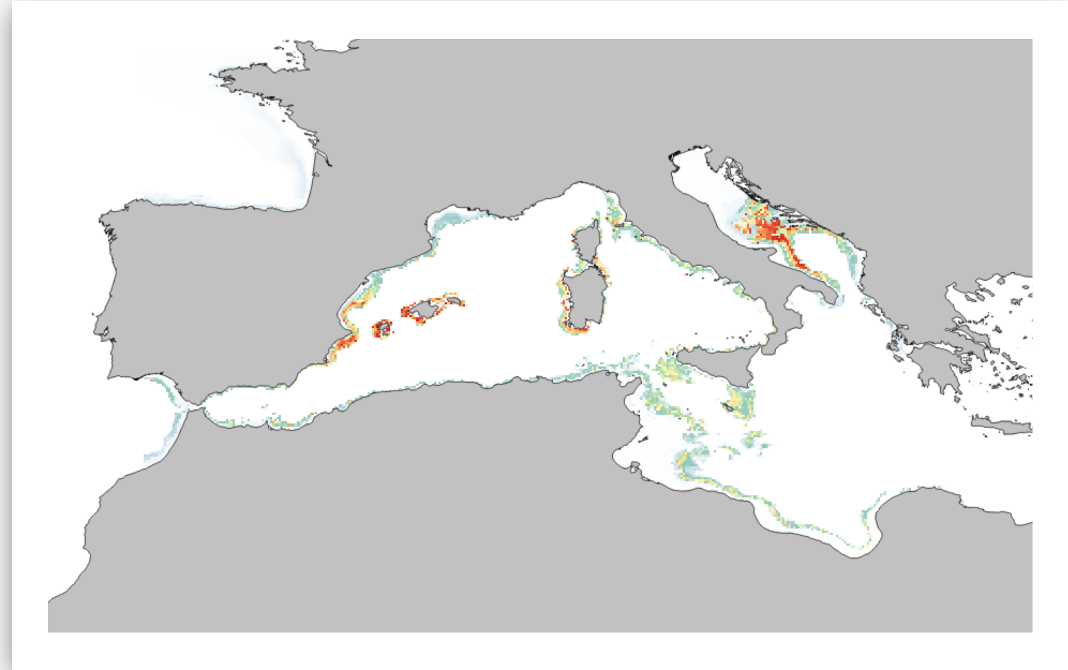

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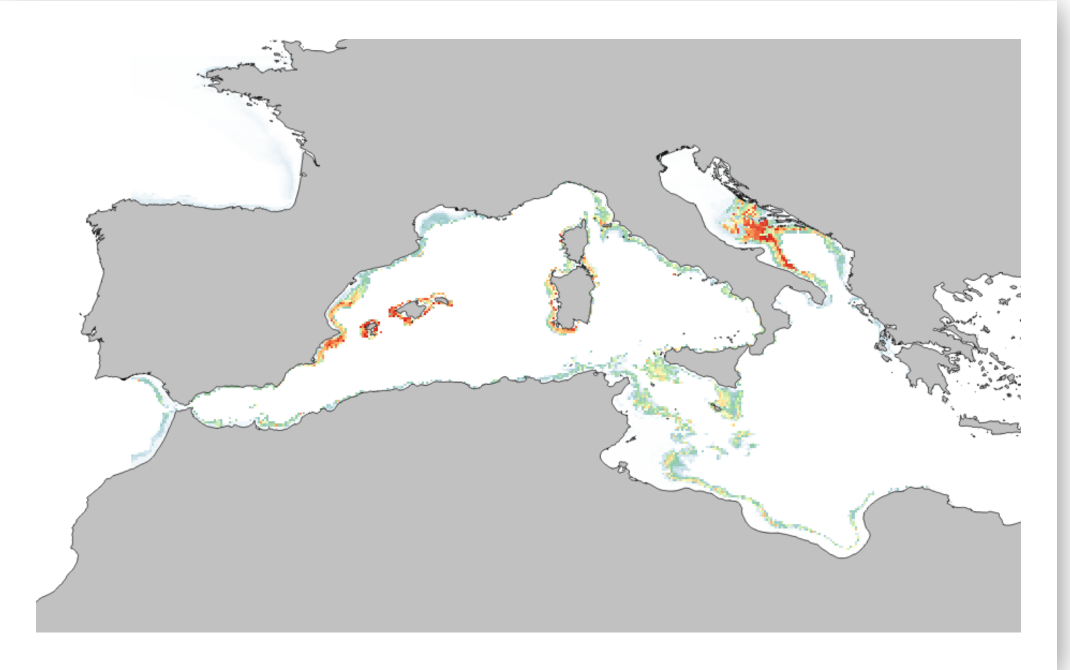

References

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Performance of algorithms in SDM

Most algorithms have been tested and compared.

Differences in performance among different algorithm types tend to be smaller than differences among species.

Machine learning algorithms, like MaxEnt, and especially those that incorporate model averaging (e.g., BRT and XGBoost) tend to have **better performance than more simple statistical methods** such as GLMs.